



WPI

Designing a Device for Monitoring Ankle Mechanics in Pickleball and Beyond

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Chapter 1: Background

1.1 Introduction

Pickleball is a growing sport that is a combination of tennis, table tennis, and badminton. Requiring the size roughly half that of a tennis court it is often played both indoors or outdoors (UCHealth, 2024). In the following sections, the background of the sport is presented, its biomechanics, different types of injuries that can occur, and the mechanisms of those injuries. Finally, we identify gaps in knowledge to enable better injury prevention when playing pickleball, which motivates this project.

1.2 Physiological Benefits of Playing Pickleball

Pickleball's popularity has grown, especially in older age groups, due to its physiological benefits and activity level. It is recommended by the U.S. Department of Health and Human Services and the CDC that adults who are older should exercise at a moderate or vigorous intensity for at least 150 minutes a week (Casper et al., 2023). According to Santos et al. (2023), it is predicted that by 2030, physical inactivity could lead to 500 million non-communicable diseases. Examples of non-communicable diseases include obesity, cardiovascular disease, hypertension, and Type 2 diabetes (Smith et al., 2018). Playing pickleball a few nights a week, for a couple of hours at a time, allows people to meet that minimum. This amount of physical activity would allow for the heart to reach increased heart rate zones, helping with cardiorespiratory fitness, fat loss, and healthy blood circulation to the muscles and heart (Stroesser et al., 2024).

A study performed in 2023 analyzed the physical activity associated with pickleball, as well as its physiological benefits. Thirty-three people wore fitness monitors for two weeks, with the study comparing the recorded data, being physical activity duration and heart rate, between days when people played pickleball vs. did not play pickleball. It was determined that during the pickleball playing sessions, people averaged 68 minutes of moderate to vigorous physical activity, while having an increased heart rate for 86 minutes, compared to 28 minutes on days when pickleball wasn't played. Within this time frame, the heart rates of the players spent a majority of the time in the cardio zone (Casper et al., 2023).

Additionally, the Fitbits recorded the number of steps people took on days when they played pickleball and days they did not. It was recorded that the average steps on pickleball playing days were 10,495 steps, while on non-pickleball days were 7,047 steps. For adults over 65, it is recommended that they walk at least 7,000 steps a day, with playing pickleball serving as a useful activity to reach this goal (Casper et al., 2023).

A study was conducted to compare the physiological differences between 30 minutes of walking to 30 minutes of playing pickleball (Denning et al., 2022). Twenty-five people wore Hexoskin biometric garments to measure their heart rate during both activities. The results determined that the participants' average heart rate was 14% higher when playing pickleball

compared to walking. Their peak heart rates were also 19% greater while playing pickleball, reaching the increased heart rate zones mentioned by Casper et al. (2023). Pickleball increased the amount of calories burned within thirty minutes of playing compared to walking by 36% (Denning et al., 2022).

Pickleball can be defined as a way to improve heart health, leading to physiological benefits mentioned above (Casper et al., 2023; Denning et al., 2022; Stroesser et al., 2024). The combination of moderate to vigorous physical activity, increased heart rate zones, steps taken, and calories burned provides clear evidence on the important role pickleball plays on the body. While the benefits of playing pickleball are numerous, there are additional risks and resultant injury to consider.

1.3 Sports Biomechanics and Racquet Sports Context

Biomechanics plays a crucial role in understanding both performance and injury prevention in racquet sports. This area of study examines how the body produces and responds to motion, utilizing joint angles, velocities, and accelerations in conjunction with ground reaction forces and joint moments. In racquet sports, rapid accelerations, lateral movements, and repetitive strokes are the major biomechanical components. These components provide insight into how efficiently a person moves and highlight the mechanisms that lead to common overuse and acute injuries.

In racquet sports such as tennis and badminton, kinematic analyses, particularly those by Kahn et al. (2004), have demonstrated that improper sequencing of trunk rotation, shoulder motion, and wrist action can lead to inefficient strokes and increased stress on the upper extremities. Pickleball, with smaller courts and lighter paddles, requires frequent short sprints, lateral shuffles, and abrupt changes in direction, which place demand on joint mobility and balance. Falls in pickleball often occur after inadequate deceleration mechanics or backward lunging, which disrupts stability and increases the risk of ankle sprains and wrist fractures. Identifying harmful kinematics is essential for developing corrective strategies, such as improving step mechanics during lunges or promoting proper trunk and hip alignment during strokes.

Kinetics analyzes the internal and external forces acting on the body during play. Ground reaction forces during lunges, decelerations, and cutting motions are relevant for pickleball, as they determine joint loading in the ankle, knee, and hip. In racquet sports such as tennis, kinetic analyses have determined that the aforementioned common movements lead to high ground reaction forces, ranging from 1.5 to 2.7 times a person's body weight, acting upon the ankle and knee, as well as high joint torques. Martin et al. (2021) also found that a defensive open-stance forehand increases vertical ground reaction force and knee torques compared to neutral stances. In general, deceleration phases in sports are characterized by high peak forces and rapid loading rates (Harper et al., 2022). Together, these findings support that rapid deceleration may contribute to tendon strain and ankle instability, especially when repeated over time.

Understanding the biomechanics of pickleball and player movements makes its physiological benefits more prominent.

1.4 Injury in Pickleball

Due to the nature of pickleball, there are a variety of injuries that can occur, based on age, gender, and play style. An analysis of recreational tennis injuries found that among all injury types to the body during play, those to the lower extremities were most common. A lower extremity injury includes those to the hip, thigh, knee, calf, ankle, and foot. These types of injuries are said to result from the nature of tennis itself, with the sudden jolts, direction changes, and terrain of the court (Kaiser et al., 2021). This similar trend is seen in the research article from Lambert et al. (2025), where injuries from badminton, tennis, and table tennis were compared. It was concluded that most of the injuries that occurred from the three sports were located in the lower extremities, including ankle sprains or specific ligament injuries (Lambert et al., 2025). As a note, injuries that were located in the upper extremities in tennis players, such as the shoulder or arm, were from overuse. In this context, overuse could be defined as playing more than three times a week, with little breaks in between, and improper warm-ups (Hanks & Myers, 2025). Within racquet sports, upper extremity injuries also include sprains or fractures to the wrist, as it is used for protection when falling. Additionally, injuries to the knees are very common in racquet sports like tennis, due to the quick change in direction. Knee injuries could include tearing of the meniscus and different ligaments, as well as an overall sprain. The jolting, or sudden change of direction is a likely cause of a knee sprain (Greiner, 2019). Injuries in the knees can also be related to improper warm-ups, and they cause the largest deficit in performance (Owoeye et al., 2025).

Greiner (2019) and Ahn et al. (2025) found that, similar to the sports above, lower extremity injuries are the most common type of injury in pickleball. Since the surface of the court is similar to tennis, and the same type of footwear is involved, it is a reasonable conclusion that tennis injuries, more specifically, lower extremity injuries, will also occur in pickleball (Greiner, 2019).

In the history of the sport, pickleball has traditionally been dominated by an older demographic, however recently there has been an explosion of younger individuals, now incurring a relatively even age spread. The data represented in the article from Ahn et al. (2025) stated that people in the age range of 61-70 years old experienced the most pickleball-related injuries. It was concluded that older players were more likely to sustain fractures than younger players, who were more likely to sustain sprains/strains. As people age, the body experiences gradual biological changes, like a decrease in muscle mass or cellular damage accumulation (*Aging*, n.d.). Along with a decrease in muscle mass, bones also tend to shrink in size and density. These two factors can cause a detrimental effect on coordination and balance, in addition to increasing the likelihood of fractures.

The twisting and lunging motions in pickleball have a higher chance of injuring an older player due to the difference in body responses. These differences can cause additional problems

between the different age groups, such as when a fall occurs. For instance, a short fall can cause some scrapes, bruises, or even a sprain in a young adult. However, that same fall for an older adult can cause more severe injuries, including fractures (*The Older We Are the Harder the Fall*, n.d.). Younger players also appear to have a faster recovery time compared to older players, causing reinjury to be less common (Ahn et al., 2025). This greater capacity to play games consecutively causes younger players to overexert themselves. Older players seem to realize when they need to take a break, leading to more upper extremity injuries in the younger age group.

Another common trend was that older female players were more likely to sustain fractures compared to older male players (Jr et al., 2023). This can be attributed to the physiological differences between women and men. Osteoporosis is a condition where the body experiences low bone density, leading to a predisposition for fractures, which tend to occur as a woman goes through menopause. The average age women reach menopause is around 50 years old, with a decrease in hormones such as estrogen becoming common. Estrogen is important for bone density, with a significant decrease being seen during the first couple of years postmenopausal (*Menopause - Symptoms and Causes*, n.d.). Since women are more likely to experience falls than men, falling becomes riskier as there is a higher chance they could obtain a fracture (Keen et al., 2025). As mentioned above, the physiological differences between pickleball players can impact the body's response to different mechanisms present in the sport.

1.5 Mechanisms of Injury

Pickleball shares movement patterns with other racquet sports, given that it was derived from a combination of tennis, badminton, and table tennis. The sport presents unique risks, particularly when it comes to slips, falls, and rapid multidirectional changes. Myers and Hanks (2024) identified lunging, backpedaling, and sudden changes of direction as the most common movement patterns linked to falls and injuries in pickleball. These motions place high demands on lower-body stability, often in players who are older and may have reduced proprioception or neuromuscular control. Quantitative analyses from racquet sports such as badminton have shown that lunge maneuvers can involve substantial joint motion and loading. For example, the hip joint angle during lunges can increase by approximately 7-8°, and peak reaction forces and ankle/knee joint moments increase during lunging, highlighting the biomechanical stress on lower extremities (Liu, 2024).

The biomechanics of slips have been studied extensively in gait research. Redfern et al. (2001) demonstrated that slips occur when the required coefficient of friction (RCOF), being the ratio of shear to normal ground reaction forces, exceeds the available friction between footwear and a surface. Slips are most likely to occur at heel strike or just before toe-off, when shear forces are at their peak. Balance recovery requires rapid neuromuscular responses, namely increased knee flexion moments, to bring the foot back under the body, with additional support from the hip extensors. If slip distances exceed 10 cm or heel velocity surpasses 0.5 m/s, recovery becomes increasingly difficult, often resulting in falls.

Environmental and human factors also contribute to fall risk. Surface roughness, compliance, and fouling alter the available friction, while individual factors such as gait patterns, vision, proprioception, and neuromuscular health influence the occurrence of slips and recovery. Visual distractions can further exacerbate these risks. Larkin (2024) noted that communication with teammates or tracking the ball may reduce focus on stability, delaying recovery responses during a slip or fall.

These mechanisms are highly relevant for pickleball, where quick accelerations, lateral shuffles, and backpedaling require precise foot placement and quick balance recovery reflexes. The overlap of biomechanical slip research and observed injury mechanisms in pickleball suggests the sport's rapid growth has overtaken the development of specific injury-prevention strategies based on biomechanics. Incorporating quantitative data from other closely related racquet sports like badminton and tennis, paired with collecting data within pickleball, can provide a basis for understanding and identifying high-risk movements to develop preventative measures for pickleball players.

1.6 Rising injury incidence and healthcare burden

Recent data indicate that pickleball has become an outlier among racquet sports in terms of injury trends. Ahn et al. (2025) found that emergency department visits related to pickleball increased from just four cases in 2012 to nearly 300 in 2022. The sport's rapid growth in popularity, especially among older adults, has been accompanied by a parallel increase in injury-related costs and hospital visits, creating new healthcare and economic concerns.

Ryan et al. (2019) emphasize that non-contact youth sports already generate millions of dollars in inpatient and emergency department expenses. This demonstrates how even non-contact sports such as pickleball impose a substantial financial burden on the healthcare system. In the context of pickleball, this financial strain is magnified by the sport's demographic profile: older adults who are more likely to require costly hospital care after an injury. Zaifman et al. (2025) project that orthopedic sports injuries in adults aged 65 and older will nearly double by 2040, highlighting how a recreational activity primarily enjoyed by seniors could amplify demand for Medicare and Medicaid-funded treatment.

Current estimates suggest that pickleball injuries may already cost the U.S. between \$250 and \$500 million annually (Jackson, 2023). Firouzbakht et al. (2025) documented a 765% increase in hand, wrist, and finger injuries over the last decade, with over half of these cases occurring from fractures due to falling. As these injuries accumulate, they divert healthcare resources and taxpayer funding that could otherwise be used toward more pressing or life-threatening conditions such as cardiovascular emergencies, cancer care, or pediatric trauma.

Although pickleball is often perceived as a lower-impact alternative to tennis, Greiner (2019) underscores that it is not injury-free. Instead, the combination of rapid growth in participation, an aging player base, and rising injury severity has created a growing public health concern. Unlike tennis, where injury rates have stabilized or declined, pickleball now represents a unique case study of how a recreational activity can impose heavy burdens on both individuals

and the healthcare system. In today's world, it is very common for hospital beds, surgical resources, and government-funded care to be consumed by sports-related injuries. These injuries could have been prevented through better awareness, training, or protective interventions to free up the healthcare system. The sharp rise in pickleball-related injuries underscores how quickly a recreational trend can evolve into a significant public health and economic burden.

1.7 Ankle Loaded Dorsiflexion

Acute traumatic injuries can occur during pickleball because of sudden pivots or turning movements (Greiner, 2019). Injuries include ankle sprains, ankle fractures, or achilles strains. Lower extremity injuries in pickleball have also been on the rise (Kingston et al., 2024). Repetitive ankle injuries like ankle sprains can lead to chronic ankle instability (CAI) (Hubbard et al., 2007). This is a bigger risk for older players as stability declines with age (Bok et al., 2013).

Ankle loaded dorsiflexion is the angle of the tibia (lower leg) going over the foot. Ankle loaded dorsiflexion can be an important measurement for many reasons, mainly how it affects lower extremity injuries (Terada et al., 2013). Rehabilitation of ankle dorsiflexion is important after acute ankle sprains. Inadequate rehabilitation can lead to future ankle sprains or other lower extremity injuries.

There is a negative correlation between ankle dorsiflexion range of motion (ROM) and ground reaction forces (GRFs) (Almansoor et al., 2023). A reduced ROM means higher GRFs, which can lead to an increased likelihood of repeat injury to the lower extremity. A reduced ROM can also mean a reduced force absorption ability in landing. Ankle dorsiflexion ROM can influence dynamic balance, the ability to shift one's center of gravity during movement, which is essential for fall mitigation (Basnett et al., 2013). The normal range for ankle dorsiflexion ROM is around 15 degrees to 30 degrees, however with low or impaired ROM, either from stagnation or injury, recovery can be prolonged leaving players in a position of increased risk.

1.8 Current Solutions

Assessment of ankle dorsiflexion ROM is most commonly performed using static clinical tools such as a handheld goniometer or inclinometer for weight-bearing lunge tests (Konor et al., 2012). During these tests, clinicians measure tibial inclination relative to the foot to determine peak dorsiflexion angle. These methods are inexpensive and easy to use, making them standard in rehabilitation settings, however they typically provide only a single end-range value and rely on manual alignment, which can introduce variability and limit repeatability.

More advanced biomechanical assessment systems, such as three-dimensional motion capture, can precisely quantify ankle joint angles and calculate displacement from neutral stance to peak loaded dorsiflexion (Knudson & Elliott, 2004). When paired with force plates, these systems can correlate the joint angles and ground reaction forces (Martin et al., 2021; Harper et al., 2022). While highly accurate, these technologies are costly, require laboratory environments, and are impractical for routine screening or frequent reassessment.

Portable alternatives such as Inertial Measurement Units (IMUs) provide an intermediate solution. When placed on the shank and foot, IMUs can estimate sagittal plane ankle motion and detect changes in joint angle during weight-bearing tasks. Although more accessible than laboratory systems, IMUs are susceptible to drift, require calibration, and are not specifically designed for isolating controlled measurements of natural stance alignment and peak loaded dorsiflexion angle change.

Given reduced dorsiflexion ROM is negatively correlated with increased ground reaction forces (Almansoor et al., 2023), accurate quantification of ankle angle under load is clinically important. Despite this, most currently available tools either provide simple static ROM values without quantifying progressive angle change from natural stance, or they require complex laboratory setup to obtain precise kinematic data.

This creates a gap between basic clinical measurement and high-level biomechanical analysis. There is a need for a practical, lower-extremity focused system capable of accurately measuring ankle alignment in natural stance and quantifying peak loaded dorsiflexion angle change in a controlled, repeatable manner. Such a solution would improve screening and rehabilitation monitoring while remaining accessible to the public.

Chapter 2: Design

2.1 Proposed Data Acquisition

Before generating a proposed design, the team first needed to identify exactly what portion of the lower extremities to focus on. As stated above, ankle injuries in pickleball players constitute much of the strain on the healthcare system as a whole. With this in consideration, the team collectively decided to focus on the ankle area, specifically around the lateral malleolus, as this location is undertracked when considering movements specific to pickleball, as well as generating useful data to develop preventative measures for injury.

After deciding on the location to collect data, the specific data to acquire was discussed. Myers and Hanks (2024), identified lunging, backpedaling, and sudden changes of direction as the most common movement patterns linked to falls and injuries in pickleball. This highlights the need to measure ankle flexion, especially ankle loaded dorsiflexion. Limited flexion can lead to sprains and knee collapse, with proper flexion supporting vertical forces and aiding stability. The measurement of ankle flexion allows for the identification of potential risks as well as areas to target for improvement.

The data that can be collected opens several promising avenues for reducing the burden on the healthcare system. This information could be used proactively, to identify specific movement patterns and body regions that may need strengthening to reduce injury risk. The data can also be collected and displayed to a specific player via a smart device to prevent high-risk movements and improve play. Collecting this metric can identify many risks associated with pickleball, and with proper measures in place, can be used to improve gameplay as well as reduce the risk of injury.

2.2 Methods of Data Collection

Once the data the team wished to measure was finalized, sensors that could generate this data were sought. Upon looking at sensors on the market, the group quickly found accelerometers to be the best choice to use for the measurement of linear acceleration. Fong et al. (2011), in a study conducted using various accelerometers on the lower extremities, stated its usefulness for the purpose, also citing the average range of acceleration read between 2-15g for most sensors. Considering 6g is considered high physical activity for an individual, it is safe to assume the sensor will function correctly. While accelerometers provide data such as pitch and roll (rotation around the y & x axes), in order to read our 3D motion, yaw (rotation around the z-axis) is required. To measure this, a gyroscope can be used in conjunction to read data on angular and linear acceleration, allowing for a full range of 3D motion.

An unfortunate side effect of intense physical activity on gyroscopes, however, is the presence of drift, which slowly skews the angles read by the sensor. When no motion is conducted, the gyroscope reads a small, nonzero value, which over time can stack and massively alter the readings. As such, it is imperative that a way of correcting this drift is included in our

collection method. The addition of an internal magnetometer will correct this through reading the Earth's magnetic field to provide heading information, acting as a compass, preventing drift from getting too large.

After determining the required sensors for our data collection, an Inertial Measurement Unit (IMU) was agreed upon as the best option. IMUs are electronic devices that contain built-in accelerometers & gyroscopes to measure roll, pitch, and yaw, with some models containing magnetometers in addition for correction, making it optimal for the design. Requiring a battery and wiring to power, it was ideal for non-invasive tracking of movement. Costing as little as \$20 and lightweight, given our intended use, research was then conducted into the best choice for design.

After some initial brainstorming with the team, a potential design set up was decided. This setup included two design components, a heel cuff housing one IMU and microcontroller, and a shank hub housing a second IMU. The heel cuff would be placed on the back of the shoe, with the IMU directly on the back of the heel, in line with the IMU on the shank. The shank hub would be attached about three quarters up the calf, once again with the IMU inline with the IMU on the heel cuff. This initial design concept can be seen below in **Figure 2.1**.

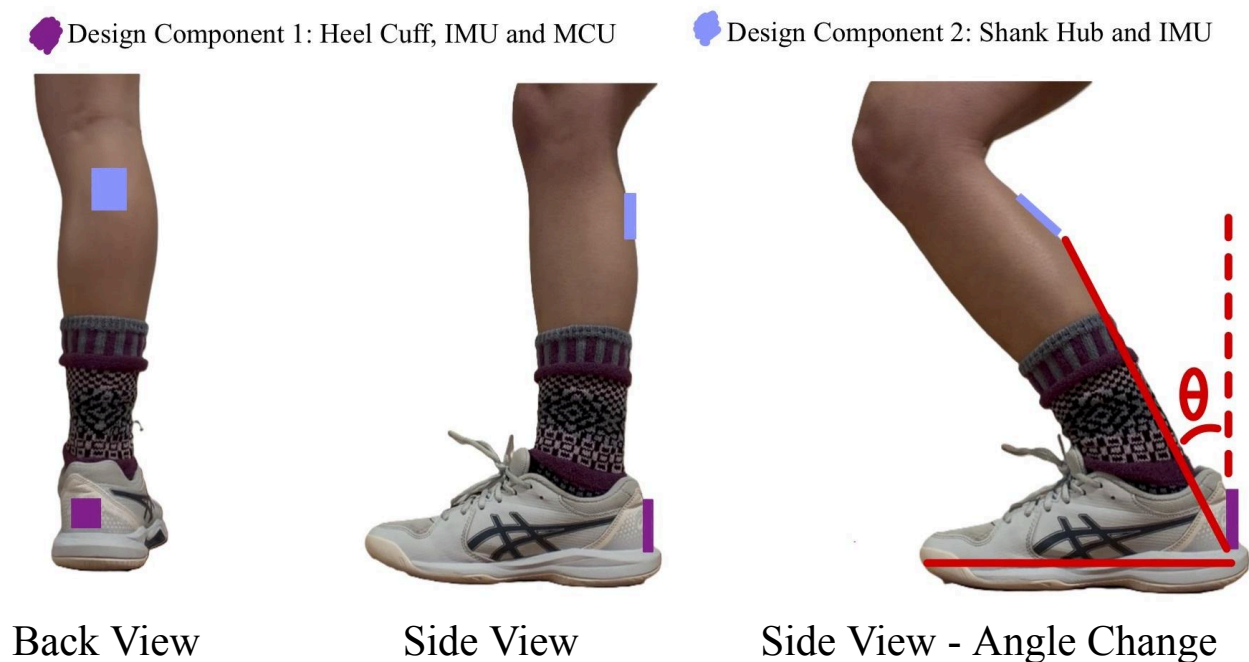


Figure 2.1 Potential two IMU system setup

2.3 Sensor Design Research

The design process began with the selection of the HUZZAH ESP32 Feather as the microcontroller. This board was chosen due to several features identified through product documentation and community forums. The board has built-in wifi and Bluetooth, enabling real-time wireless streaming of motion data, which is vital to our project. The board supports Arduino IDE and has extensive community project examples and reference coding projects.

Most importantly, the board has a strong reputation for powering multi-sensor projects, whether it be multiple sensors of the same type or different types. This aligned with the intended two IMU architecture. At the time, the ESP32 appeared to be a strong and flexible foundation for a wearable motion tracking system.

After assembling the system, which consisted of the ESP32 board and two BNO085 IMUs, the group was unable to establish any sort of connection to the IMUs through even a basic I2C scanner pulled from manufacturing documentation. Despite correct wiring, neither IMU appeared on the bus. Subsequent review of technical forums and Adafruit documentation revealed a critical issue that had initially been overlooked. The BNO08x family is highly demanding and unstable on ESP32 I²C buses. The sensor uses large SHTP packet transfers and strict timing requirements that conflict with the ESP32's I²C implementation. Numerous documented cases confirm that even when detected, packet loss and communication freezes occur under polling. Due to this, continuing with the ESP32 would require abandoning the BNO085 sensor that is well-suited for ankle motion tracking.

The BNO085 is one of the highest-performing IMUs available for wearable motion analysis, confirmed across Adafruit documentation, community research, and academic projects. One such project is the HillcrestLabs BNO085 Motion Tracking Solution for VR by Ceva, Inc., where a BNO085, paired with an external reference camera, was used to completely mitigate drift within a VR headset. Unlike most IMUs that only output raw accelerometer and gyroscope data, the BNO085 was specifically chosen due to its providing:

- Best-in-class drift-free sensor fusion
- Absolute orientation
- 9-DOF orientation
- Nearly zero drift over a minute

For ankle biomechanics, absolute orientation is required to compute reliable joint angles because the fusion occurs on a chip. Therefore, keeping the BNO085 became a non-negotiable design requirement, prompting the switch away from the ESP32.

After confirming that the ESP32 was fundamentally incompatible with the BNO085, the microcontroller was switched to the Adafruit Feather Sense. The BNO08x family of sensors was listed as compatible on the Adafruit documentation and contained all of the same features as the original ESP32 board selected. The feather sense (nRF52840) boasts a stable and well-supported I2C implementation, has no known communication issues with the BNO08x family, includes an onboard 9-DOF IMU (LSM6DS33 / LSM6DS3TR-C), and maintains the BLE (Bluetooth low energy feature).

Several approaches were attempted to operate two external BNO085 units simultaneously using the Feather Sense board, but problems persisted. The BNO085 supports only two addresses: 0x4A and 0x4B. Both ship with address 0x4A, creating an immediate conflict. Three strategies were attempted.

Attempt #1: LTC4316 I²C Address Translator:

The LTC4316 was purchased to artificially shift one BNO085's address. However, Adafruit specifically outlines that the BNO085 does not work with address translators, something that was missed upon reading the documentation. The sensor's SHTP protocol breaks under translated addressing. This method was not viable.

Attempt #2: TCA9548A I²C Multiplexer:

The next approach was to isolate each IMU on separate I²C channels via a TCA9548A mux. This theoretically eliminates address conflicts because each IMU occupies a different branch. In testing, this resulted in inconsistent IMU detection and failure to pull data from more than one IMU at a time. After research into community forums and Adafruit documentation, it was found that the BNO08x family does not work reliably with muxes. During community forum research, it was found that one user was able to perform a similar project using two separate libraries.

This final workaround was attempted using two separate SparkFun BNO08x libraries, one using the older SH-2 protocol and one using the newer version. However, both libraries define conflicting SH-2 structures and expect exclusive control of SHTP. This attempt, while potentially possible, was abandoned as the primary objective of this project is the wearable device, and this would have pulled the project toward a CS project.

Attempt #3: Manual Address Change

While researching community forums to attempt to get the multiplexer to work, a thread in a forum pointed out that the IMUs have an option to manually switch their address. Research suggested that tying the DI (SA0) pin to 3.3V could force a BNO085 to use the alternate address 0x4B. This method did work, and I²C scanning confirmed detection at 0x4A and 0x4B simultaneously. However, this resulted in the final discovery that the Adafruit BNO08x library cannot handle two active SHTP sessions on the same I²C bus, even with unique addresses. This essentially validated community reports that two BNO085 sensors cannot reliably coexist on a shared I²C bus.

Attempt #4: Setup Configuration Change

During the troubleshooting process, the team reassessed the system configuration and evaluated whether dual-IMU sensing was necessary. The Adafruit Feather Sense (nRF52840) demonstrated reliable communication with a single external BNO085 IMU. Upon further review, it was recognized that the BNO085, as a 9-DOF sensor with onboard sensor fusion, independently provides stable yaw, pitch, and roll orientation outputs without requiring a second IMU for orientation computation.

This realization enabled simplification of the system architecture. The final configuration consists of the Feather Sense microcontroller and a single external BNO085 connected via

STEMMA QT. This configuration eliminates I²C address conflicts and ensures stable, repeatable sensor operation.

With the IMU positioned on the shank, the measured angle represents the change in tibial orientation relative to the planted foot during dorsiflexion. Because ankle dorsiflexion in a weight-bearing condition primarily reflects tibial progression over a fixed foot, relative orientation change from a neutral baseline provides an appropriate representation of ankle motion. As long as the IMU produces consistent orientation change measurements, ankle flexion can be accurately quantified.

2.4 Device Design Process

Due to the changes in the sensor progression, the team changed the design set up. Two design components are still utilized, but the heel cuff houses one microcontroller, and the shank hub houses the IMU. The placement is the same as the potential two IMU system setup seen in **Figure 2.1**. The final design set up is demonstrated in **Figure 2.2**.

Design Component 1: Heel Cuff and MCU

Design Component 2: Shank Hub and IMU



Back View



Side View



Side View - Angle Change

Figure 2.2 Proposed Device Setup

2.4.1 Heel Cuff Design Process

Table 2.1 Pairwise Comparison Chart for objectives identified during background research

Objective	User Friendly	Lightweight	Comfortable	Durable	Cost	Total Score	Multiplier for Pugh Analysis
User Friendly	x	1	1	1	1	4	5
Lightweight	0	x	0	1	1	2	3
Comfortable	0	1	x	1	1	3	4
Durable	0	0	0	x	1	1	2
Cost	0	0	0	0	x	0	1

Before beginning the design stages, objectives for the device were generated and ranked. When deciding on which objectives to prioritize, considerations were made based on points such as device function/objective, testing parameters, time wearing, and location. As such, the most important aspect to consider for the design was deemed to be user friendliness, or rather, the ease of operation and function of the device for the wearer. This is especially important for a variety of reasons, from allowing the wearer to understand the data being collected, to the clarity and accuracy of data, as well as a more consumer-based viewpoint, saving time and improving efficiency for anyone who may own the device. Next, considering consumer preferences towards wearable devices, a comfortable design is also mandatory. With little to no comfort, players will have lower motivation to wear the device during play. In addition, a light weight is important for our design considerations. A heavier housing system will impair play, however the expected effect is slight. Comparing comfort to weight, it was determined that a more comfortable design is desired, and players feel more inclined to wear it as opposed to the slight compromise in weight for our specific design case. While this may decrease the total number of players using the device, the ease of use and ability to collect accurate data were deemed more essential. Lastly, with the intense physical movements involved in pickleball, a durable device is needed. Prolonging device life will help with data collection and provide more useful information to players.

Design 1: Arm System



Figure 2.3 Image of the 3D model of Design 1: Arm System

This preliminary design is an early prototype intended to securely mount the ankle-tracking electronics housing to a user's shoe during motion-capture testing via tightening arms, seen in **Figure 2.3**. The fixture consists of a compact sensor enclosure paired with two arms that wrap around the shoe for stability. The arms have a curved geometry that conforms to the shoe's sidewalls, improving fit. The enclosure has space for the onboard electronics and includes mounting threads for thumbscrews, which allow the arms to be tightened and locked into position. This design prioritizes fit and ease of adjustment.

Design 2: Shoelace Hole

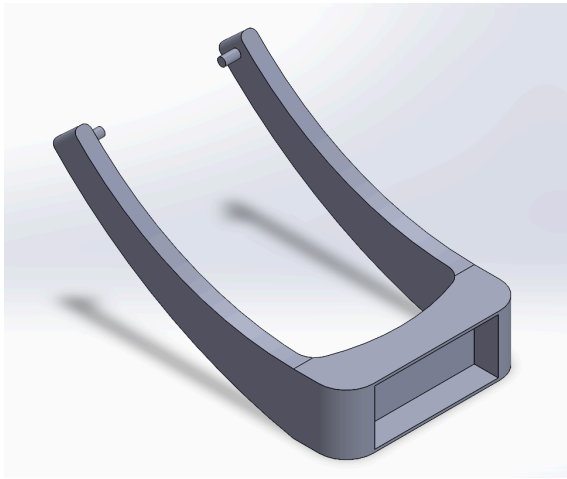


Figure 2.4 Image of the 3D model of Design 2: Shoelace Hole



Figure 2.5 Image of pickleball shoe with the extra shoelace hole circled, demonstrating location of attachment for Design 2: Shoelace Hole

This design includes pin holes to fit into the unused shoelace holes to keep the hub in place, also allowing for a user-friendly attachment, seen in **Figures 2.4 and 2.5**. On the back of the design there is room for Feather Sense with additional space for any future sensors. In order for the device to fit a wide range of users, the arms of the design would be able to be adjusted using a sliding and locking mechanism. The material would be made out of plastic or PLA from a 3D printer, allowing for the device to be lightweight. There is no connection between the user's foot and the device, allowing for a comfortable fit.

Design 3: Force Fit

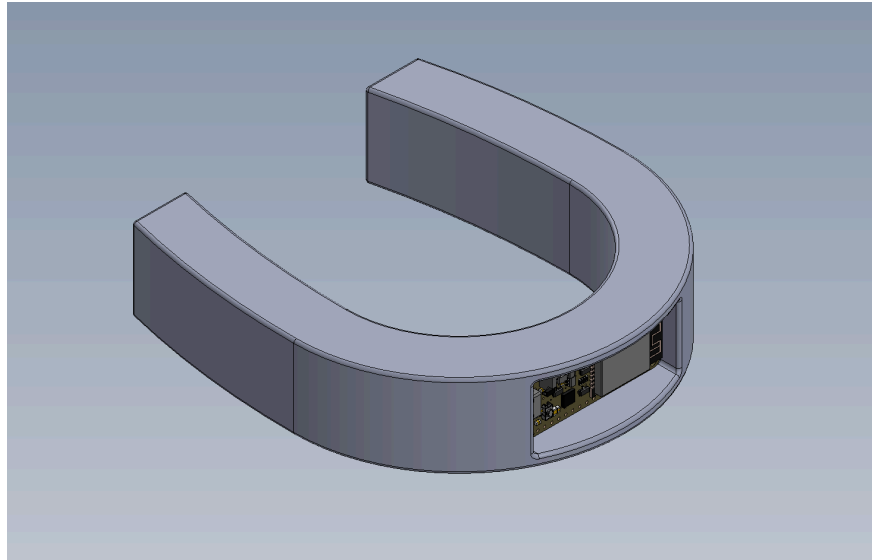


Figure 2.6 Image of the 3D model of Design 3: Force Fit

This initial design concept features a force-fit heel cuff intended to house all the necessary sensors on the back for ankle kinematic data collection used for injury prevention. The U-shaped cuff wraps securely around the heel, providing a snug fit without the need for additional fasteners, which improves ease of use and comfort during pickleball play, as seen in **Figure 2.6**. This design prioritizes minimal interference with movement while maintaining sensor stability, one of the key concepts, as the device undergoes further design iteration.

Table 2.2 Pugh Analysis Chart for determining design choice

Criteria	Multiplier	Arm Clip	Shoelace Hole	Force Fit
User Friendly	5	-1	0	0
Lightweight	3	-1	+1	+1
Comfortable	4	-1	-1	0
Durable	2	0	-1	0
Cost	1	-1	-1	-1
Total	X	-13	-4	2

Rules:

0: if less important

1: if more important

-1: if equally important

To move forward in the design process, a Pugh Analysis chart was used. 5 different criteria were identified and then ranked by importance. These criteria were defined as important things to include in our final design. Each design was given points based on those criteria and in the end the force fit design was the best option of the three.

Design Iterations

The initial redesign in **Figure 2.7** established the fundamental shape and structure of the cuff. At this stage, the geometry focused on ensuring general compatibility with the ankle. The design prioritized basic stability, however the cuff was too big, fitting loosely around a shoe, with the edge of the cuff sitting too far up on the shoe near the arch. Flexibility proved to be difficult, and further material-specific optimization was required. Slots for the ESP32, BNO085, and 3.7V 500mAh battery are provided, from left to right.

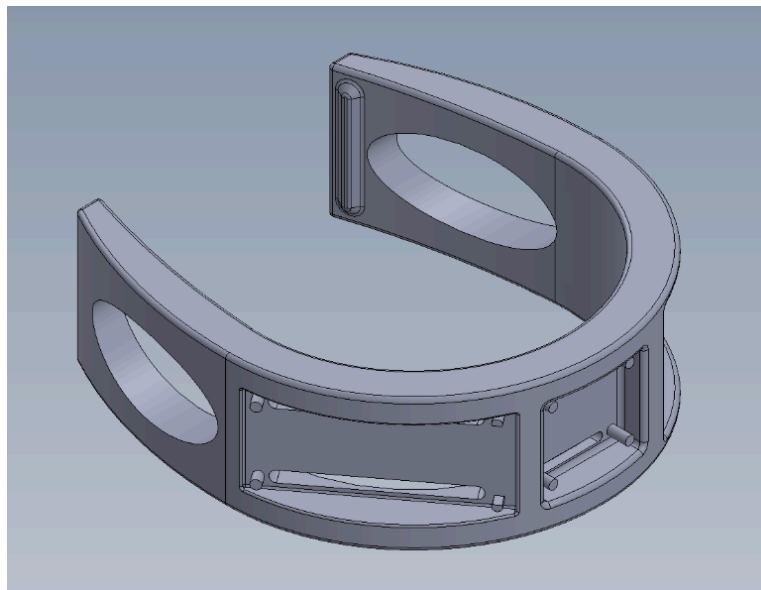


Figure 2.7: V1.0 - Baseline Redesign

In **Figure 2.8**, the opening of the cuff was shifted to create flexible, pitch-fork-like arms. This was to allow the cuff to slide on more easily and grip onto the shoe, ideally performing better at accommodating a wider range of foot sizes. From V1.0, the length of the arms was reduced to grip around the heel just enough to hold the cuff in place. The thickness of the cuff was reduced as it was unnecessary material, in turn improving comfort and decreasing weight of the cuff to be more wearable. The one thing that was lacking was that the design, being entirely PLA, did not allow the cuff to conform to the foot when putting it on.

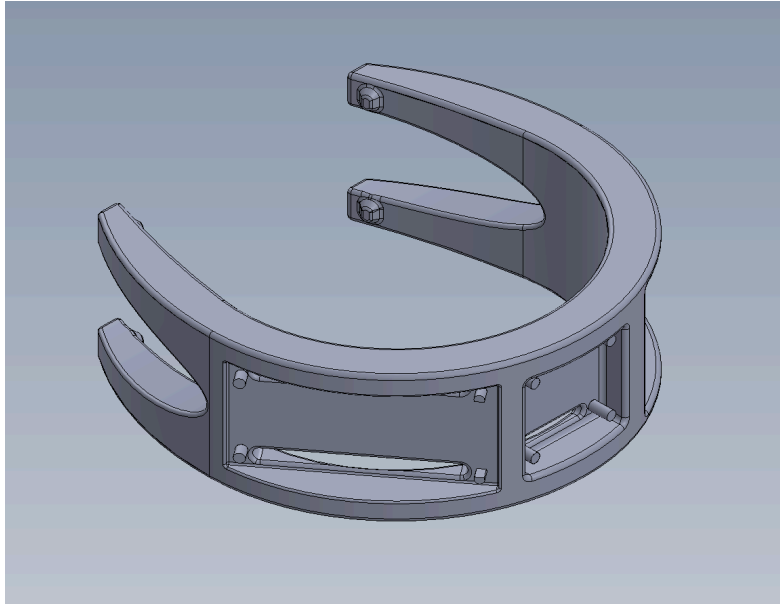


Figure 2.8: V1.1 - Flexible Arm Geometry

Building on the flexible arm concept in **Figure 2.9**, V1.2 separated the cuff arms into distinct components, connected to the base by pin-to-hole connections, to take advantage of dual-material printing. TPU was used for the arms to provide elasticity for better conformity, while PLA formed the rigid base for sensor mounting and structural support. This separation allowed the design to balance comfort, flexibility, and mechanical stability while keeping the overall weight minimal.

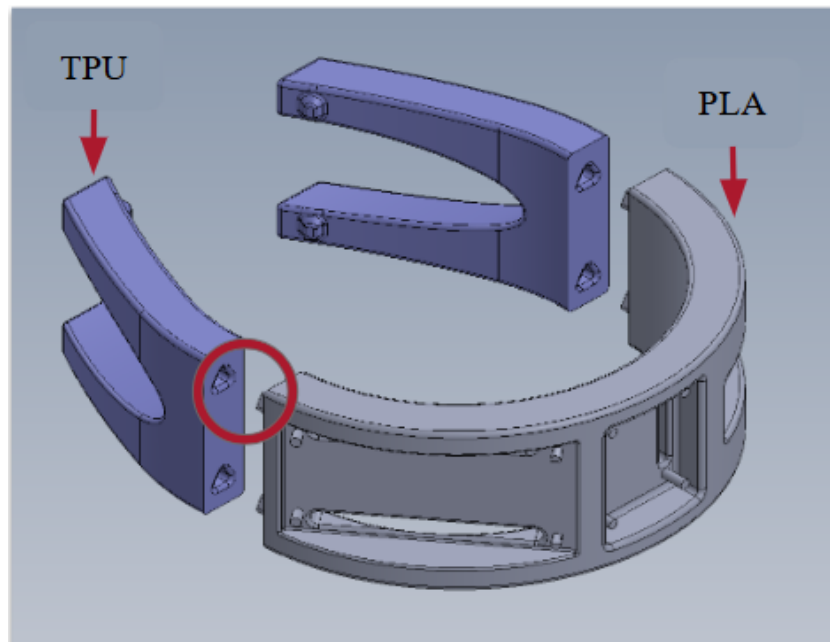


Figure 2.9: V1.2 - Dual Material Separation

The iteration in **Figure 2.10** refined the internal geometry to be more in line with the heel-like structure on a shoe. Slots were adjusted to accommodate the Feather Sense, now located on the back. The Feather Sense features an area dedicated to the onboarding IMU and another area for the 3.7V 100 mAh battery, positioned to the left. Ports were added to facilitate mounting the sensors and testing the device. The mounting area for the Feather Sense was extended back so that the length of the cuff could be reduced, removing excess material, with the thickness of the cuff reduced as a result of reducing components on the back. Given TPU's flexibility, the arms were able to be additionally tapered to enhance grip on the shoe. This print successfully fits a range of shoe sizes. One issue that persisted is the separation of the TPU arms from the PLA back, causing a gap formation due to the pin and hole connections being internal.

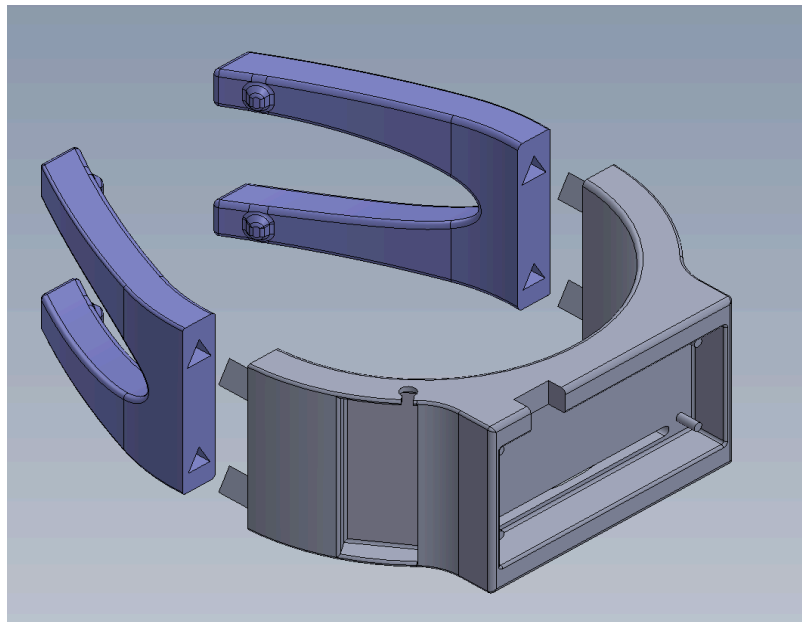


Figure 2.10: V1.3 - Optimized Fit + Wiring

For **Figure 2.11**, V1.4 focused on securely integrating the Bluetooth module. A dedicated internal slot was integrated to house the Bluetooth hardware, ensuring it stays protected and stable during movement. There was a reversal of the connection geometry between the PLA housing and the TPU arms. By moving to a PLA-hole and TPU-slot configuration, the design utilizes the rigidity of the PLA back, allowing the PLA rivets to seat securely within the flexible TPU. This change provides the necessary give to eliminate the material separation and gaps that occurred in previous versions during arms flexion in order to accommodate wider shoes. While this version significantly improves the durability of the joint, future iterations will need to address the final wire routing paths and potential long-term fatigue of the TPU slots under constant high-flexion use.

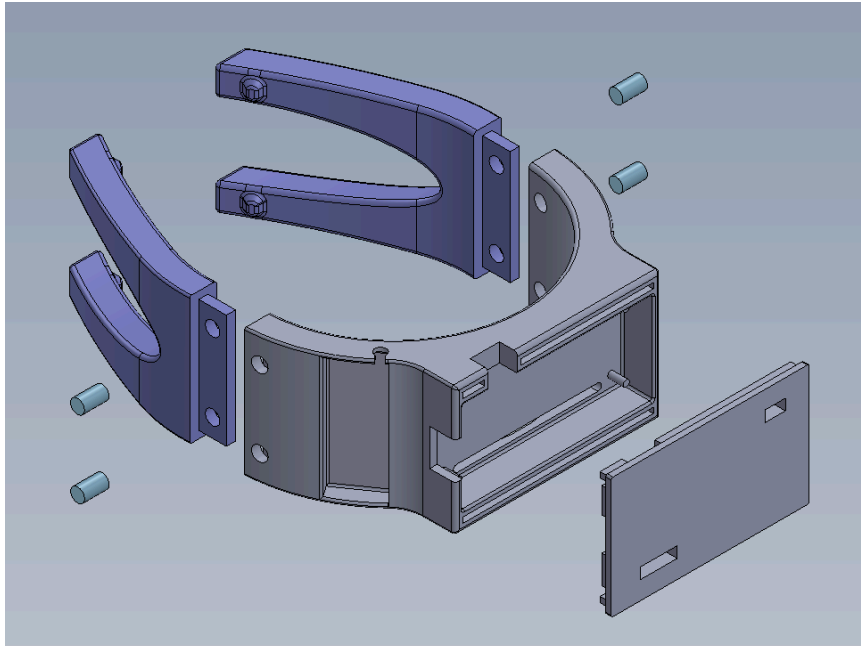


Figure 2.11: V1.4 - Secured Connection

3D Printing Materials

Design iterations V1.0-V1.1 were originally 3D printed using PLA filament, however, after printing, the group realized that switching to TPU would be more flexible and durable than PLA. Design iteration V1.1 was printed in TPU, but it was found to be too flexible, removing the force fit aspect from the design. This resulted in the decision to print the physical housing in PLA with the arms printed in TPU, seen in design iteration V1.2. It was determined that the back of the heel cuff should continue to be printed in PLA to have a tighter fit to the back of the shoe. Meanwhile, TPU allows for the arms of the design to have more flexibility. When sliding the cuff onto the shoe, the arms are pointed more inwards, but the TPU allows the arms to widen depending on the size of the shoe, while the PLA keeps the force fit.

As an additional note, for the shank hub, each design iteration was printed in PLA for stability and strength.

2.4.2 Shank Hub Design Process

The first Shank Hub design consisted of a small housing space, 1.1" x 1.2", where the IMU could sit, seen in **Figure 2.12**. Within the square, there were four pins that aligned with the four pin holes on the IMU in order to keep the IMU in place. The pins were extruded from wider yet shorter cylinders to allow the IMU to sit elevated, allowing a strap to be fed through the openings on both sides of the housing, allowing for attachment to the back of the shank. Additionally, a port was added to the bottom of the housing to allow wired IMU connection to the Feather Sense.

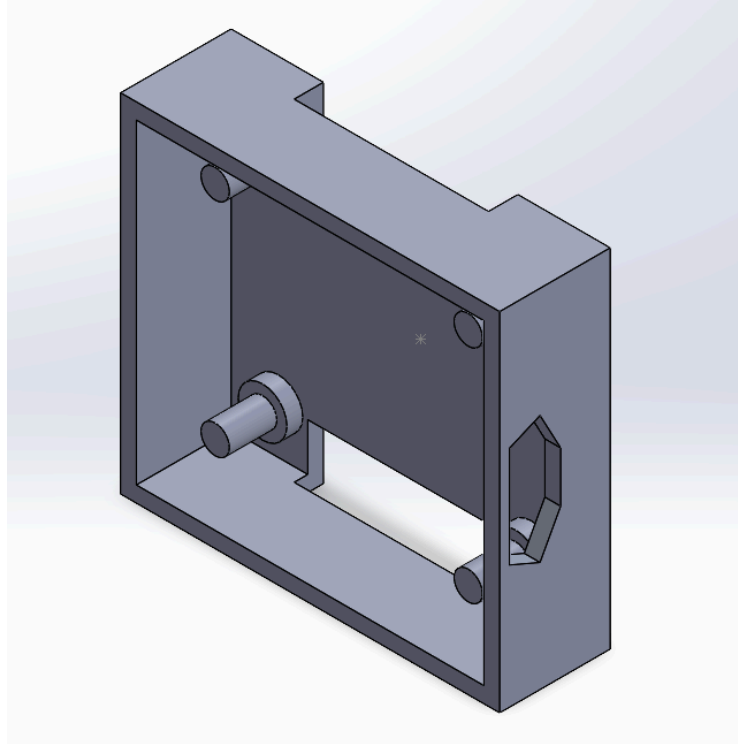


Figure 2.12: V1.0 - Prototype

As stated above in our earlier sensor setup, it was originally assumed a Translator board was required to run a two IMU setup. With this assumption, a space for the translator was added to the initial Shank Hub design, described above, seen in **Figure 2.13**. Similarly to the first design, openings were added to each side of the housing to allow for straps to be fed beneath the sensors and through the housing to attach to the shin. Within this design, two straps would be needed for shin attachment.

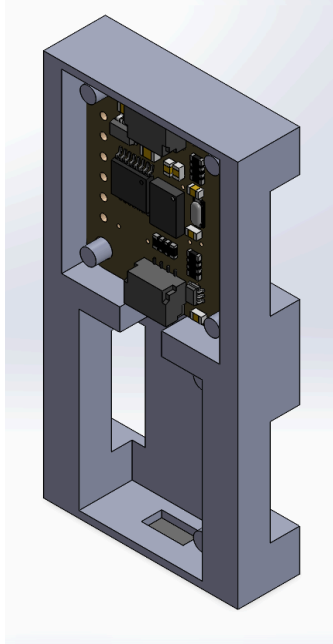


Figure 2.13: V1.1 - Prototype & Translator

After initial design prints, the housing was deemed too cumbersome to wear accurately. In order to add comfort during play and a more snug fit to the leg, a small curve was added to better shape the shank, seen in **Figure 2.14**. Instead of attaching the housing via straps, a sleeve would be slipped over the housing, similarly to a soccer shin guard. The sleeve would be made out of athletic material to discourage deterioration over time due to sweat.

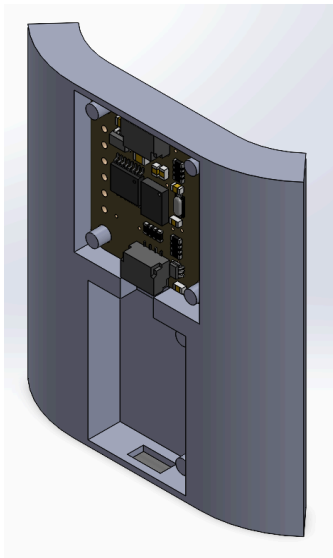


Figure 2.14: V1.2 - Curved Fit

Due to the removal of the translator during additional sensor design and research, a new design was now needed, focused solely on the IMU. A fourth housing design for the second IMU sensor was developed to be placed on the lower back end of the shank, around 1.1" x 1.2" x 0.35", seen in **Figure 2.15**. A hole was cut for the wiring connection between the IMU and Featherboard housed on the shoe, with openings for a $\frac{3}{4}$ " elastic velcro strap to hold in place. A lid was developed to enclose the IMU to prevent falling out, affixed via 8 extruded pins embedded 0.035" into the snap lid. Due to the small size, a box shape was deemed acceptable, as opposed to a curve fit.

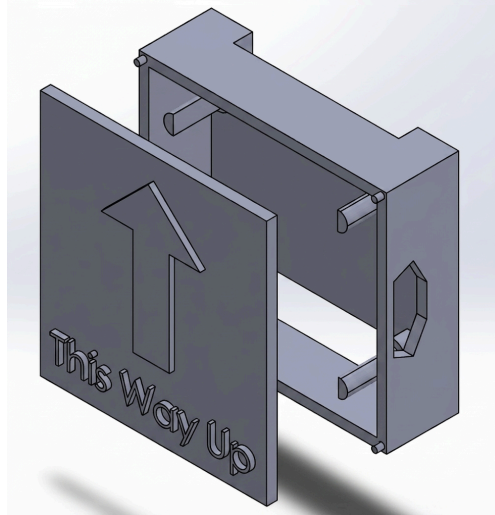


Figure 2.15: Version 1.3 - Strap Alignment + Lid

After this design was complete, the team began receiving the materials required to operate the design, such as the IMU and velcro strap. Through this, the need for more changes was highlighted in **Figure 2.16**. The provided velcro strap was wider than originally assumed, requiring an increase in cutout width from $\frac{3}{4}$ " to $\frac{4}{8}$ " to account. In addition, the 8-pin lid design did not hold the IMU in place efficiently, as any weight applied would cause the lid to pop off. A new clip-like design was implemented to better affix the housing lid to base. A directional arrow was also included for orientation assistance.

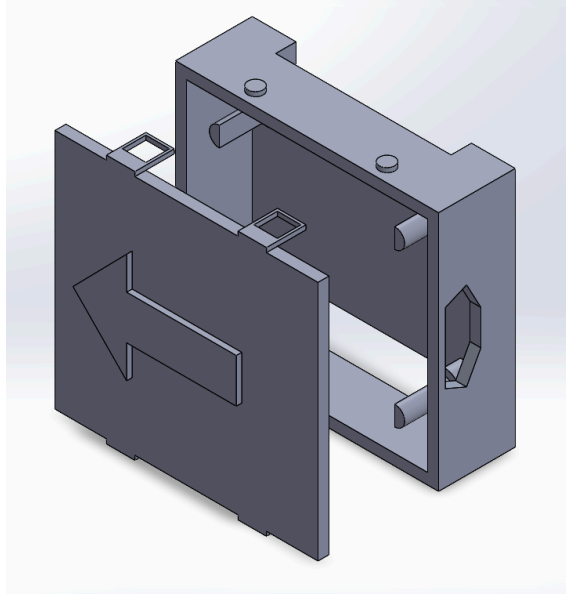


Figure 2.16: Version 2.0 - Redesigned Lid

After printing of Version 2.1, a new problem arose, that being the abilities of our printer to produce the designed clips consistently. After experimenting with differing sizes and thicknesses, the team decided to pivot to a new lid adhesion design, focusing on a through pin force fit as seen in **Figure 2.17**. The thickness of walls was increased to 1.3" x 1.4" , allowing for wider indents as well as better structural stability of the housing overall.

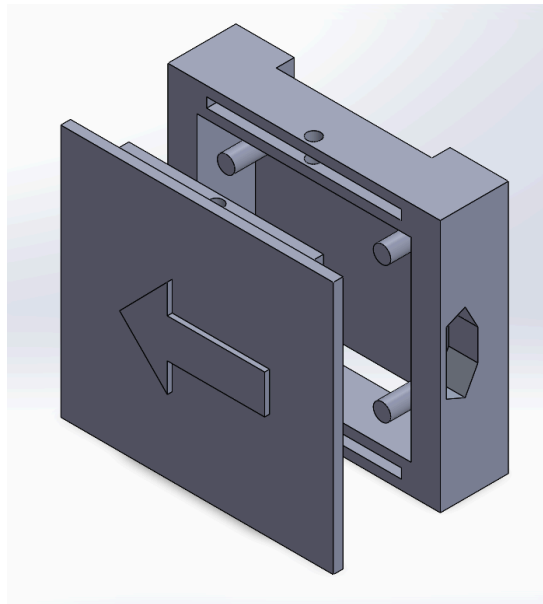


Figure 2.17: Version 2.1 - Through Pin

After printing Version 2.2, the team was overall satisfied with the operation of the housing. Testing began at this stage, however more quality of life improvements were identified, the first being the orientation of the IMU. Due to the location of the connection port, the IMU was orientated in a manner that read pitch during dorsiflexion, as opposed to the roll intended. To rectify this, the port and velcro straps were rotated 90° in **Figure 2.18**. Secondly, the IMU was extremely difficult to remove from its housing. With the size of the housing overall being relatively snug to the IMU itself, efforts to grab or shake the IMU out were ineffective. To make removal easier without compromising the snug fit of the housing, a 0.15” hole was added to the base of the housing, allowing for pressured removal from the bottom. The intent was to aid in “popping” the IMU out of the housing without interfering with the more delicate electronics on top.

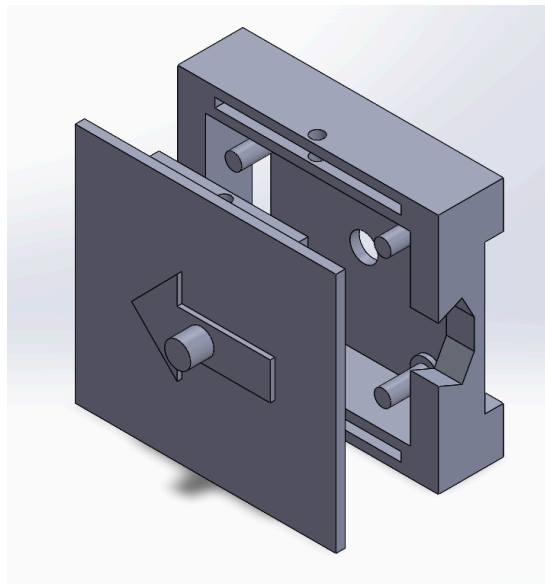


Figure 2.18: Version 2.2 - Orientation & QOL Improvements

Due to the orientation change from Version 2.3, one problem arose that needed to be rectified. Due to the velcro strap cut being located on the 1.3” side from the 1.4”, the cut bore too deeply into the IMU housing pegs, and as such could no longer support the IMU while stable. A new design was required to allow for strap insertion without compromising the structural integrity of the 4-peg system. After consideration, an increase in height for the housing was added, adding more area to the bottom of the housing allowing for a less intrusive cutout, in turn allowing for a stable strap connection and completely removing the need to cut through any housing pin. As it stands currently, this is our final design in **Figure 2.19**.

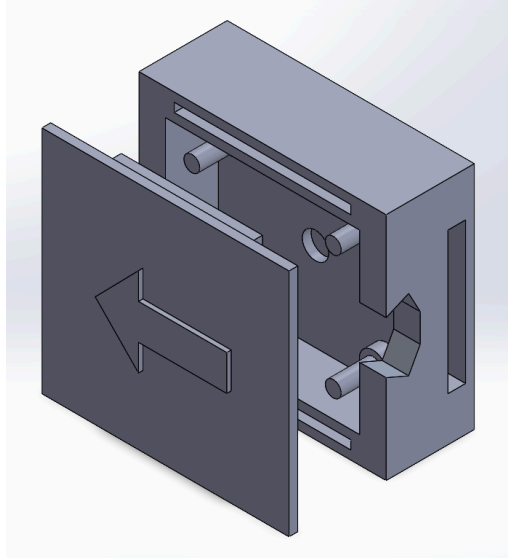


Figure 2.19: Version 2.3 - Structural Stability

Chapter 3: Measurement Methods and Testing Protocols

3.1 Methods Overview

This chapter describes the procedures used to collect and analyze ankle dorsiflexion data using the wearable sensor system. First, the process used to acquire raw IMU data is described. Next, the method used to convert this raw data into meaningful ankle angle measurements using MATLAB is presented. The accuracy of the IMU-based measurement system is then validated against sagittal plane video measurements obtained using Kinovea motion analysis software. Finally, the participant testing protocol, range of motion (ROM) calculation, and repeatability analysis methods are described.

The primary outcome measured in this study is ankle dorsiflexion ROM, defined as the angular change between a neutral stance and the peak tibial progression over the ankle during a controlled forward lean.

3.2 Raw Data Collection

Data acquisition is performed using a wearable sensor system consisting of a BNO085 IMU mounted on the shank hub and connected to the heel cuff microcontroller. The IMU measures orientation using onboard accelerometer, gyroscope, and magnetometer sensors combined through internal sensor fusion algorithms. Orientation data is recorded as Euler angles (roll, pitch, and yaw) at a sampling rate of approximately 100 Hz. The raw dataset consists of time-series measurements containing timestamps and corresponding orientation angles in degrees. This data is exported for further processing in MATLAB.

Data will be collected from a diverse group of pickleball players varying in age, gender, and skill level. Each participant will provide measurements of both right and left ankles. The wearable system records the relative tibia over ankle angle with respect to the foot. The raw data collected by the shank hub IMU will be processed through MATLAB using filtering algorithms to compute the joint angle.

The following variables will be recorded:

- Peak dorsiflexion angle at the end of flexion
- Neutral stance

These values will be used to calculate variation of dorsiflexion for a limb and right vs. left limb symmetry then compared against established healthy ranges of dorsiflexion reported in literature. Limited dorsiflexion has been associated with increased risk of ankle sprains. Symmetry between limbs has also been an indication of compensatory movement patterns caused by instability. By identifying participants whose dorsiflexion values fall below the accepted safe ranges or who demonstrate significant symmetry between limbs, they can be alerted as higher injury risk and given recommendations to strengthen areas of weakness. Additionally, the device can track dorsiflexion improvements during physical therapy or

rehabilitation. Repeated measurements over time allow the wearer, physicians and physical therapists to monitor recovery progression and inform them of what recovery methods are most effective.

3.3 Angle Measurement and Data Processing

The raw data from the IMU must be processed to extract useful ankle dorsiflexion measurements. This processing is performed using a custom MATLAB graphical user interface (GUI) developed for this project.

The MATLAB program imports the raw orientation data recorded during each trial and applies filtering algorithms to remove sensor noise and small fluctuations caused by vibration or body sway. A low-pass filter is applied to the roll angle signal to isolate the smooth forward tibial rotation associated with dorsiflexion movement.

Once filtered, the program identifies two key points within each trial:

Neutral Angle (θ_{neutral}) - The tibial angle recorded while the participant stands upright in a relaxed position prior to movement.

Peak Angle (θ_{peak}) - The maximum forward tibial rotation recorded when the participant reaches their maximum dorsiflexion position.

Since the IMU is mounted on the back of the calf and rotated approximately 90° from its origin orientation, the neutral angle typically has a larger value than the peak angle during forward movement. The MATLAB GUI automatically detects these values and exports the peak and neutral angles for further ROM analysis described in Section 3.6.

The software then stores the calculated ROM value for each trial and compiles the results across all trials for further analysis. The MATLAB GUI also provides visualization tools that allow researchers to inspect the roll angle time series and confirm that the correct peak and neutral points were identified. A sample output of the MATLAB processing interface is shown in Figure X. This automated processing simplifies ankle angle measurements calculations and ensures consistency across all trials and participants.

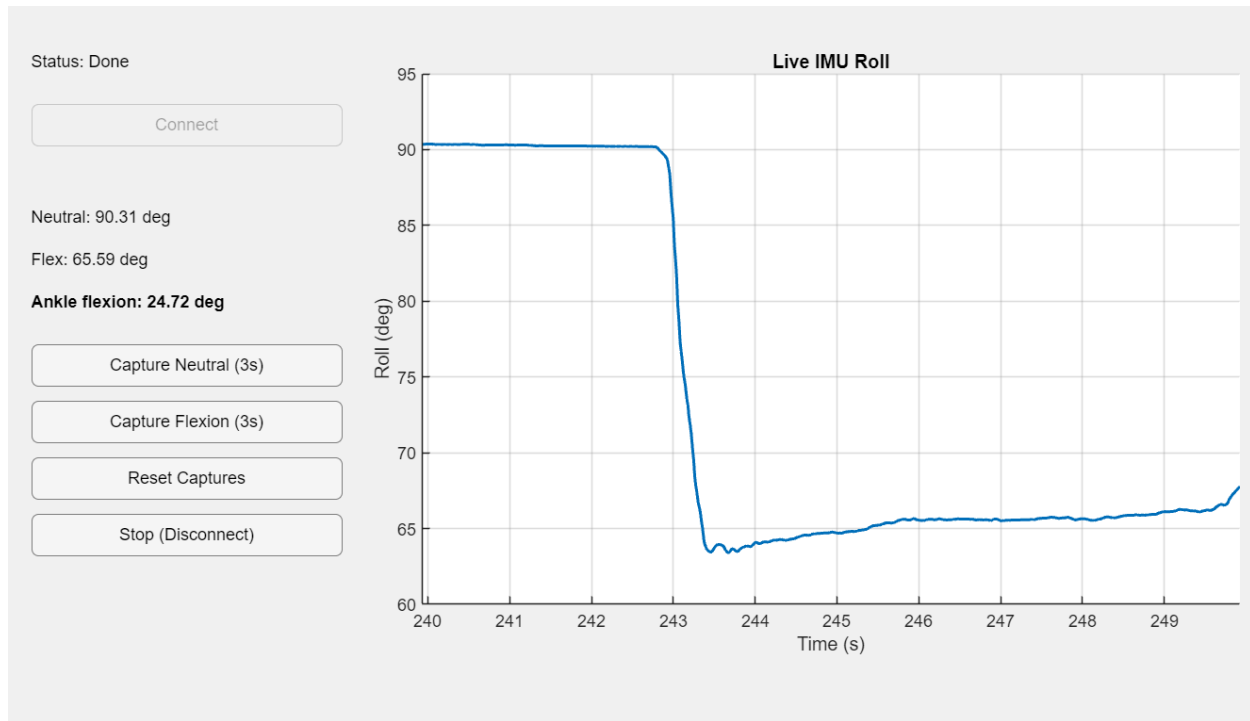


Figure 3.1: Example image of the MATLAB GUI.

3.4 Measurement Validation

To evaluate measurement accuracy, the IMU derived ankle dorsiflexion values were compared to sagittal plane video measurements obtained using the Kinovea motion analysis software. Kinovea is a widely used open-source motion analysis tool commonly used in sports biomechanics and rehabilitation research to measure joint angles from video recordings. The software allows users to manually define anatomical landmarks and calculate angles between body segments using frame by frame analysis. Because Kinovea is an open source program and is easy to use, it was selected as the reference measurement method to validate the wearable sensor data.

During validation trials, ankle dorsiflexion movements were simultaneously recorded using both the wearable sensor system and a video camera positioned perpendicular to the sagittal plane of motion. The camera captured the participant's lower leg and foot during the dorsiflexion movement. Video recordings were imported into Kinovea, where the ankle dorsiflexion was found during the two movements in each trial, static standing and peak bending movement.

Peak dorsiflexion values obtained from Kinovea were paired with the corresponding values recorded by the sensor system for comparison. The first two trials of each foot were chosen to compare in Kinovea. From the data, there was an average of a 1.75 degree difference between the IMU angle and Kinovea angle.

One limitation with this method was dealing with the movement during the trials. During some movements, the subject was moving side to side and picking when the peak dorsiflexion

occurred was harder. Another limitation was the angle of the camera, during some trials the perspective of the foot was not completely straight and that might have changed the ankle angle slightly.

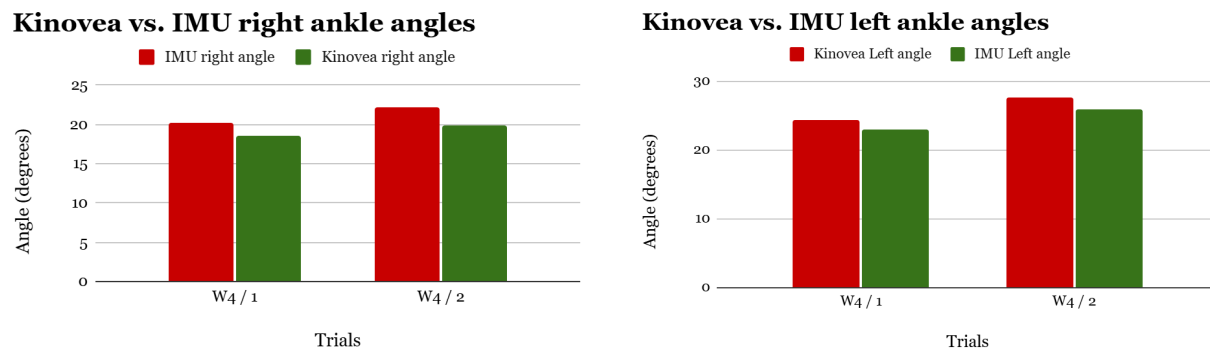


Figure 3.2: Comparison of IMU-derived dorsiflexion ROM and Kinovea video measurements

3.5 Test Protocol

The objective of this study is to quantify the tibia over ankle angle from a diverse group of pickleball players by simulating common movement in pickleball using a wearable sensor system. The system measures dorsiflexion in the sagittal plane to evaluate range of motion, symmetry, and potential injury risk factors.

Participants will wear the heel cuff device during controlled movement tasks. The device consists of an Inertial Measurement Unit (IMU) mounted along the shank hub and a micro controller unit that sits on the heel cuff. By ensuring that both are aligned longitudinally across the tibia and foot, the change in angle can be calculated easily without needing complex trigonometry. This configuration improves consistency and accuracy in measurement while reducing computational errors compared to the IMU and microcontroller randomly placed on the user.

Each participant will follow a standardized testing protocol, including:

- Static standing calibration
- Controlled dorsiflexion movement of the participants choosing

The protocol is as follows:

1. Heel cuff and shank hub are placed on the participant
2. Static standing angle is captured
3. Participant leans forward over their foot as far as they can without their heel lifting up, capturing peak dorsiflexion angle
4. Repeat five times
5. Repeat process with other foot



Figure 3.3: Image (left) depicting the static standing angle position, and image (right) showing the peak dorsiflexion angle position

As a note, participants were allowed to reach peak dorsiflexion angle using their preferred forward lean technique, but should not include leaning to the side and bringing their heel up, as this would skew the resulting dorsiflexion angle.

Data will be collected for both limbs for comparison within a subject. The primary outcome is the measure of peak dorsiflexion angle while in a planted position. Secondary measures include symmetry between limbs and variability across trials. All tests will be performed indoors on a flat court surface to replicate typical pickleball playing conditions.

3.6 Range of Motion Calculation

Ankle loaded dorsiflexion was calculated using the roll output from the BNO085 IMU positioned on the calf. The roll angle represents tibial orientation change with respect to the sagittal plane. Due to the fact that weight-bearing dorsiflexion primarily reflects tibial progression over a fixed foot, change in tibial roll angle relative to the baseline provides a valid representation of ankle motion.

For each trial, a neutral baseline (θ_{neutral}) was recorded during a relaxed upright stance. The angle recorded at the peak of the users dorsiflexion motion was identified as the peak angle (θ_{peak}).

Dorsiflexion ROM was computed as:

$$(\theta_{\text{ROM}}) = (\theta_{\text{neutral}}) - (\theta_{\text{peak}})$$

Due to the orientation of the IMU the starting neutral angle would be larger than the peak angle as the IMU is positioned on the back of the calf rotated $\sim 90^\circ$ from its origin position.

Five trials were collected per ankle. The mean ROM for each ankle was calculated per each test subject.

3.7 Repeatability Analysis

Measurement repeatability was evaluated using the standard deviation (SD) of the five trials collected for each ankle.

For each ankle of every subject:

$$SD = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

x_i = each individual trial data ROM value

$\bar{x}$ = mean ROM across five trials

n = number of data points in the data set

This data was normalized using the coefficient of variance that was calculated as:

$$CV = \frac{SD}{\bar{x}} \times 100$$

Lower SD and CV values indicate greater measurement consistency. The standard deviation is presented in units of degrees while the coefficient of variance is a unitless value that represents the data set's dispersion as a percentage of the mean.

Chapter 4: Hypotheses-driven Data Analyses

4.1 Data Analyses Overview

After establishing the accuracy and repeatability of the measurement method, the collected data were used to evaluate biomechanical characteristics of ankle dorsiflexion within the study population (IRB approved). The data analysis focused on two primary questions: 1) Whether significant symmetry exists between left and right ankle dorsiflexion ROM within the individual, and 2) Whether dorsiflexion ROM significantly differs between any of the sample groups.

4.2 Limb Symmetry Analysis

Limb symmetry was assessed by comparing the mean dorsiflexion ROM values between the right and left ankles of each participant.

To present this data as a normalized measure of difference the percent symmetry was also calculated as:

$$\text{Symmetry \%} = \frac{|\overline{\theta_{ROM,R}} - \overline{\theta_{ROM,L}}|}{\text{maximum}(\overline{\theta_{ROM,R}}, \overline{\theta_{ROM,L}})} \times 100$$

The symmetry value is normalized against whichever ankle had the larger mean ROM. This results in a value that quantifies the deficit magnitude between the subject's two ankles. If the subject's right ankle has a larger mean ROM than the resulting symmetry percentage value will be the percent difference of mean ROM between the weaker and stronger ankle. In the case of the right ankle having a larger mean ROM the resulting data will allow a statement to be made such as the left ankle side is X% lower than the right side.

4.3 Population Level Comparison

To evaluate the two questions mentioned in the overview, two hypotheses were tested using statistical methods. These hypotheses assessed potential differences in dorsiflexion ROM between limbs of an individual group and between the ROM data of the sample groups.

Hypothesis 1: Limb Symmetry

A paired samples t-test was performed to determine whether a statistically significant difference exists between the dorsiflexion ROM of the left and right ankles within a sample group.

Null Hypothesis (H_0):

There is no statistically significant difference between the left and right ankle dorsiflexion ROM.

Alternative Hypothesis (H_1):

There is a statistically significant difference between left and right ankle dorsiflexion ROM.

Paired samples t-test:

$$t = \frac{\overline{x_{diff}}}{\left(\frac{s_{diff}}{\sqrt{n}}\right)}$$

$\overline{x_{diff}}$ = mean of the differences between paired observations

s_{diff} = standard deviation of the differences

n = sample size

Since the measurements are paired within the same participant, a paired t-test was used to evaluate whether the mean difference between limbs differs significantly from zero.

Hypothesis 2: Sex Based Differences in Dorsiflexion ROM

To evaluate whether ankle dorsiflexion ROM differs between any of the four sample groups (55+ Men, 55+ Women, College Men, College Women), a one-way ANOVA was performed, followed by a Tukey-Kramer post-hoc test to identify specific between-group differences. A two-way ANOVA was subsequently conducted to decompose the observed group differences into the independent and combined contributions of age and sex.

Null Hypothesis (H₀):

There is no statistically significant difference in ankle dorsiflexion ROM between any of the groups.

Alternative Hypothesis (H₁):

There is a statistically significant difference in ankle dorsiflexion ROM between any of the groups.

The one-way ANOVA F-statistic follows the formula:

$$F = MS_{\text{between}} / MS_{\text{within}}$$

Where the MS_{between} is the mean square variance attributed to differences between groups and MS_{within} is the mean square variance within groups. A significant F-statistic indicates that at least one group mean differs from the others, with the Tukey-Kramer post-hoc test used to identify which specific pairwise comparisons reach significance.

Statistical significance was defined as: $\alpha = 0.05$

To reduce the likelihood of a type 1 error occurring (false positive), a Bonferroni correction was applied to account for multiple comparisons, adjusting the significance threshold to $\alpha = 0.05/6$ or $\alpha = 0.00833$ where 6 is the number of comparisons being made. All statistical analyses were conducted using MATLAB.

4.4 Sample Size Considerations

A target sample size of 60 participants was selected to provide the necessary statistical power for detecting differences in dorsiflexion ROM while remaining feasible within the project timeline. This sample size allows for:

- Adequate representation of male and female participants
- Capture natural variability in dorsiflexion mobility
- Reliable statistical comparison between limbs and sex groups

Pilot data from 10 participants were used to estimate effect size for power analysis. Cohen's d was calculated using the pooled standard deviation of male and female ROM values.

4.4.1 Preliminary Sample Size Estimate

An initial estimate of sample size can be calculated using the following relationship:

$$n = \frac{p(1-p) * Z^2}{E^2}$$

where:

- n = sample size
- p = effect size
- Z = z-value (determined by precision level)
- E = margin of error

This equation provides a general estimate of the number of participants needed to achieve the desired statistical precision.

4.4.2 Effect Size Estimation from Pilot Data

Pilot data from 10 participants were used to estimate the expected effect size for dorsiflexion ROM differences between sexes.

For each participant, the average dorsiflexion angle was calculated as:

$$DF_{avg} = \frac{DF_L + DF_R}{2}$$

where:

- DF_L = left ankle dorsiflexion ROM
- DF_R = right ankle dorsiflexion ROM

Participants were then separated into male and female groups, and the pooled standard deviation was calculated. Cohen's d effect size was computed as:

$$d = \frac{\bar{x}_f - \bar{x}_m}{SD_{pooled}}$$

where:

$\bar{x}_f$ = mean dorsiflexion ROM of female participants

$\bar{x}_m$ = mean dorsiflexion ROM of male participants

SD_{pooled} = pooled standard deviation of the two groups

Table 4.1: Cohen's d values based on comparisons between each group

Group 1	Group 2	Cohen's d (magnitude)
Men 55+	Women 55+	0.564
Men 55+	College Men	3.456
Men 55+	College Women	1.679
Women 55+	College Men	3.783
Women 55+	College Women	2.077
College Men	College Women	1.691

4.4.3 Sample Size for Independent t-Test

To determine the number of participants required to detect sex-based differences in dorsiflexion ROM, the sample size for an independent samples t-test was calculated as:

$$n = \frac{2(Z_{\alpha/2} + Z_{\beta})^2}{d^2}$$

where:

n = sample size per group

d = estimated effect size (Cohen's d)

$Z_{\alpha/2}$ = z-value corresponding to the significance level

Z_{β} = z-value corresponding to statistical power

For this study:

- Significance level: $\alpha = 0.00833$
- Statistical power: 0.80

- $Z_{\alpha/2} = 1.96$
- $Z_{\beta} = 0.84$

Table 4.2: Sample size per group determined with Cohen's d

Group 1	Group 2	n
Men 55+	Women 55+	50
Men 55+	College Men	2
Men 55+	College Women	6
Women 55+	College Men	2
Women 55+	College Women	4
College Men	College Women	6

From **Table 4.2**, the highest value of n was 50, but due to the feasibility of this study, that value was ignored as that is too many people to collect data from within a short time frame. Instead, the second highest n value of 6 was utilized to determine the sample size of the groups. In the end, the team used an n value of 15 to obtain more significance.

Chapter 5: Results

5.1 Range of Motion Results

Peak dorsiflexion range of motion was calculated for each participant and averaged across five trials per ankle. This data was then plotted into a box plot to capture the range of mean ankle dorsiflexion ROM within a group and compared against the other sample groups. Mean left and right ankle ROM values for each group are presented in Figure 5.1.

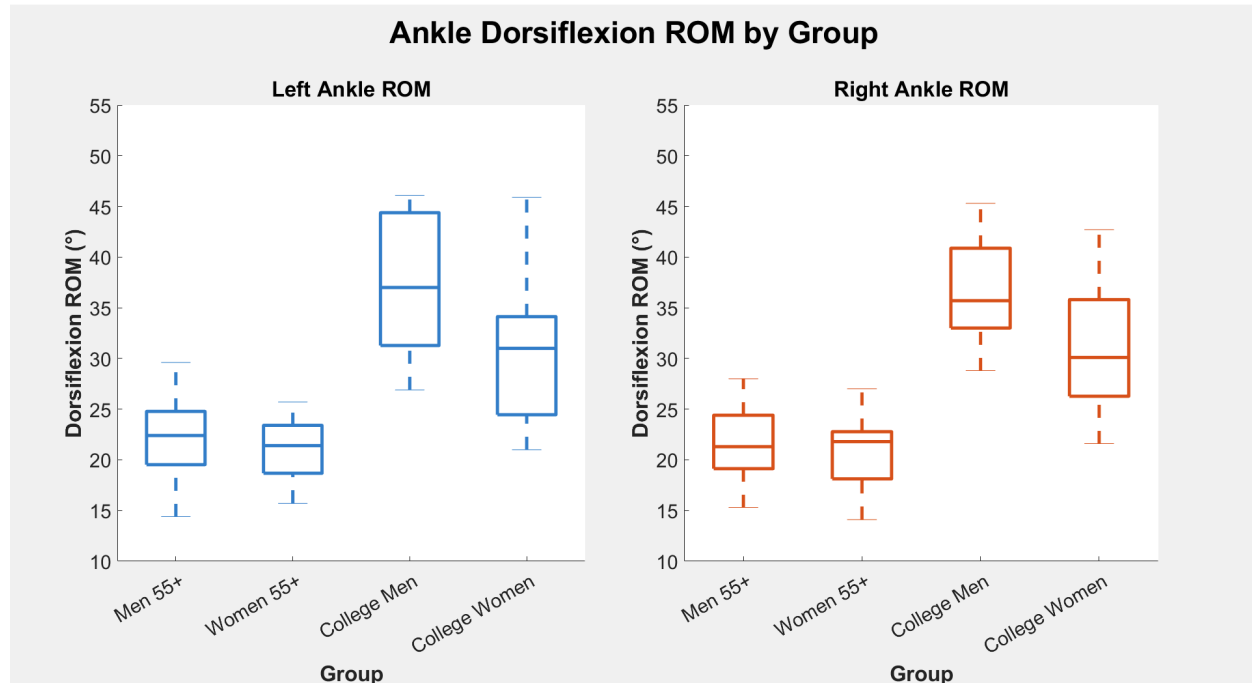


Figure 5.1: Mean ankle dorsiflexion range of motion (ROM) for each participant. Error bars represent ± 1 standard deviation across five trials per ankle.

Mean dorsiflexion range of motion values across Men 55+ ranged from 14.4° to 29.6° with a group mean of 22.3° for the left foot and 21.6° for the right foot. The highest mean dorsiflexion ROM was observed in Participant 9 (29.6° $\pm$ 3.0°), while the lowest mean ROM was observed in Participant 13 (14.4° $\pm$ 1.7°).

Mean dorsiflexion range of motion values across Women 55+ ranged from 14.1° to 27.0° with a group mean of 21.3° for the left foot and 20.7° for the right foot. The highest mean dorsiflexion ROM was observed in Participant 8 (27.0° $\pm$ 1.5°), while the lowest mean ROM was observed in Participant 5 (14.1° $\pm$ 1.5°).

Mean dorsiflexion range of motion values across College Men ranged from 26.9° to 46.1° with a group mean of 37.8° for the left foot and 36.7° for the right foot. The highest mean dorsiflexion ROM was observed in Participant 6 (46.1° $\pm$ 1.2°), while the lowest mean ROM was observed in Participant 15 (26.9° $\pm$ 2.6°).

Mean dorsiflexion range of motion values across College Women ranged from 21.0° to 45.9° with a group mean of 30.6° for the left foot and 30.7° for the right foot. The highest mean dorsiflexion ROM was observed in Participant 14 ($45.9^\circ \pm 3.2^\circ$), while the lowest mean ROM was observed in Participant 7 ($21.0^\circ \pm 1.3^\circ$).

5.2 Repeatability Analysis

Trial to trial repeatability was assessed using the coefficient of variance (CV) calculated across five trials per ankle for each participant within the four groups.

Table 5.1: Coefficient of variation (%) of ankle dorsiflexion for all four groups

	Left			Right		
Group	Min	Max	Mean	Min	Max	Mean
Men 55+	2.01	15.26	6.98	1.89	11.11	5.10
Women 55+	3.68	11.27	7.58	3.48	10.64	6.88
College Men	1.59	11.04	5.90	2.24	13.37	6.70
College Women	2.16	18.93	9.15	1.67	17.51	7.26

From **Table 5.1**, it can be seen that for the left foot, the minimum CV was seen in the College Men, with a value of 1.59%, and the maximum was in the College Women with 18.93%. Within the right foot, the minimum CV was 1.67% in College Women, and the maximum CV also occurred in College Women with a value of 17.51%. For the left foot, College Men had the lowest mean CV with 5.90%, while College Women had the largest mean CV with 9.15%. Men 55+ had the lowest mean CV within the right foot with 5.10%, and College Women had the largest mean with 7.26%.

5.3 Limb Symmetry Analysis

Inter-limb symmetry percentage was evaluated using the absolute difference in mean dorsiflexion ROM between left and right ankles for each participant, divided by the max ROM observed (across both left and right ankles) for that individual.

Group symmetry percentage values are illustrated in Table 5.2 while individual symmetry percentage values are displayed in Figure 5.3.

Table 5.2: ROM and Symmetry Percentage for each Sample Group

Group	Mean ROM (°)	Min ROM (°)	Max ROM (°)	Mean Symmetry (%)	Min Symmetry (%)	Max Symmetry (%)
Men 55+	21.94	14.40	29.60	10.86	0.77	32.61
Women 55+	20.98	14.10	27.00	9.67	2.13	25.68
College Men	37.28	26.90	46.10	8.98	2.21	20.72
College Women	30.67	21.00	45.90	11.17	3.10	26.06

Men 55+ demonstrated the widest range of symmetry values, having the lowest minimum symmetry (0.77%) and the highest maximum symmetry (32.61%). Mean symmetry percentage ranged from 8.98% (College Men) to 11.17% (College Women), representing a difference of 2.19 percentage points between the lowest and highest mean values.

Comparing age-related differences within sex groups, men showed an increase in mean symmetry percentage from college-aged (8.98%) to 55+ (10.86%), while women showed a decrease from college-aged (11.17%) to 55+ (9.67%).

5.4 Population Level Comparison

Hypothesis 1: Limb Symmetry

A paired samples t-test was performed to compare mean dorsiflexion ROM between left and right ankles within each group and result in:

Table 5.3: P-values for each Sample Group

Group	P-value
Men 55+	0.5844
Women 55+	0.6402
College Men	0.8866
College Women	0.8760

Table 5.3 displays the p-values resulting from the difference in ROM between left and right ankles. Within each group, the p-value was greater than $\alpha = 0.008833$, causing the null

hypothesis to not be rejected, as there was no statistically significant difference observed between left and right ankle ROM.

Hypothesis 2: Sex Based Differences in Dorsiflexion ROM

The independent samples t-test used to compare mean dorsiflexion ROM between male and female participants resulted in:

Table 5.4: Mean ROM Differences Between Groups and their P-values

	Young Women	Young Men	Women 55+	Men 55+
Young Women	-----	7.662° (0.0013)	8.634° (0.0003)	-7.674° (0.0013)
Young Men	-7.662° (0.0013)	-----	-16.297° (0.0000)	-15.337° (0.0000)
Women 55+	-8.634° (0.0003)	16.297° (0.0000)	-----	0.960° (0.9235)
Men 55+	7.674° (0.0013)	15.337° (0.0000)	-0.960 (0.9235)	-----

Within **Table 5.4**, A p-value of 0.008833 was utilized to determine if the differences between the groups were statistically significant. When comparing ROM differences, one group across the top was compared to the second group along the side. The results show that the comparison between each group had a p-value less than 0.00833, except for between the groups Women 55+ and Men 55+, with a value of 0.9235. The differences in ROM between these groups were statistically significant, but Women 55+ and Men 55+ did not present statistically significant differences in ROM.

5.4.1 Statistical Assumptions

Normality of the data was assessed using Q-Q plots for each group in **Figure 5.2**. Visual inspection revealed that most datasets were close to normal distributions, with data points falling close to the reference line. Minor deviations from normality were observed in some groups, most notably in the College Men left foot data, which showed slight deviation in the lower tail of the distribution.

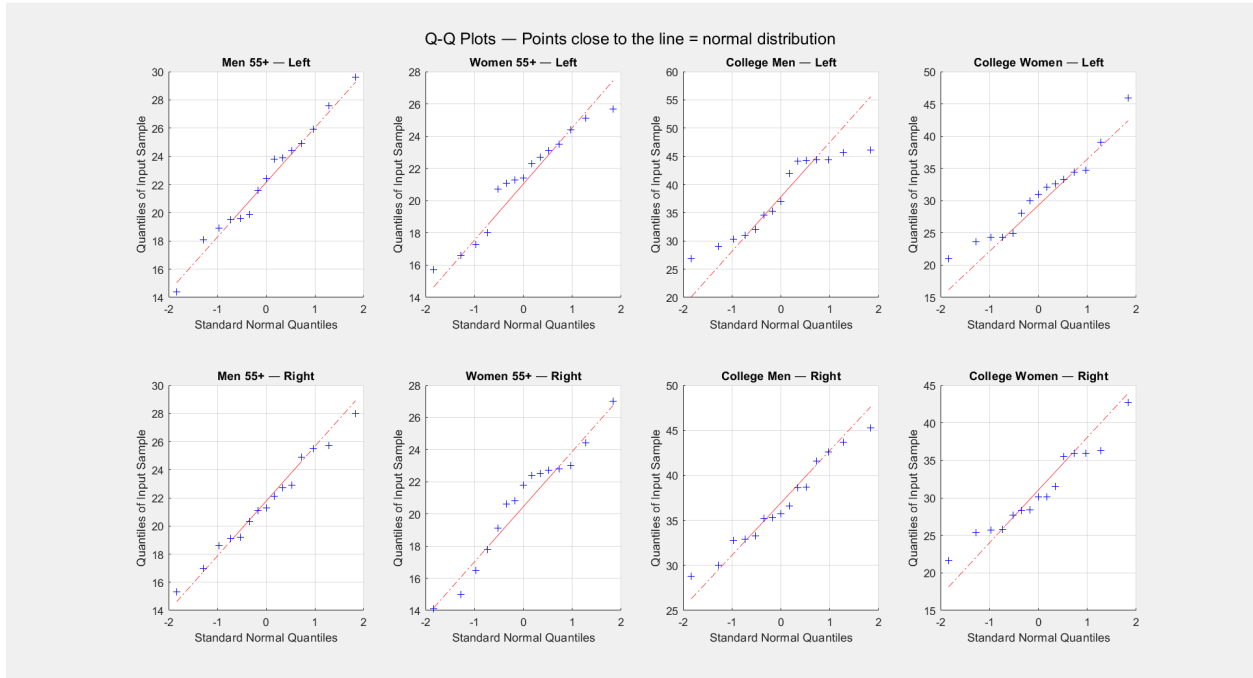


Figure 5.2: Normality testing of each foot for each sample group

These deviations from perfect normality can be attributed to the small sample size ($n=15$ per group), which limits the ability to fully capture the underlying population distribution. Additionally, the use of mean values averaged across five trials per participant reduces the impact of individual measurement variability and produces more normally distributed data according to the Central Limit Theorem.

Given the minor nature of the observed deviations, the absence of extreme outliers, and the relative soundness of the t-tests to moderate discrepancies from normality with sample sizes of $n \geq 15$, parametric testing was deemed appropriate for all analyses.

Chapter 6: Discussion

6.1 Interpretation of ROM results

Per literature review, the normal range for ankle dorsiflexion ROM is around 15° to 30°, with lower ankle dorsiflexion putting individuals at risk of injury due to tears and falls (CDC, 2023). With an absolute minimum observed dorsiflexion ROM of 14.1 from Women 55+ and 14.4 from Men 55+, all other individual data was either within or above this threshold. As this data exceeds the projected numbers dictated by literature, it can be assumed the participants are above the threshold for immediate risk. These results are consistent with expectations with both ankles.

6.2 Measurement Repeatability

When measuring ankle angle dorsiflexion, the reliability of the data must be verified to ensure accurate and precise readings, and as such, analysis must be conducted. Calculating Standard deviation (SD) and Coefficient of Variation (CV) allows the team to do just this, giving an estimate of precision around the mean to predict accurate measurements and methods.

Especially in the case of rehabilitation monitoring, precision is paramount. With a small ROM change resulting from a healthy to hazardous ROM, the precision of the measurement device must be confirmed. If this is not the case, individuals have the potential to become falsely cleared and deemed within a safe range, resulting in further decrease in ROM and injury.

CV was calculated over each batch of 5 trials for each participant in every age group. Both left and right ankle CV was calculated separately. From these calculations, the average CV values observed across all groups over both ankles ranged from 5.10% from the right ankle of Men 55+, and 9.15 from the left ankle of College Women. With a mean CV value calculated to be lower than 10%, we can assume there is a high consistency within the data.

6.3 Limb Symmetry Implications

Ankle symmetry is present in most individuals, with ankle flexion often varying off of several factors including uneven usage or preference for the dominant foot, previous injury and inconsistent stretching. Compensation rehabilitation screening, a process of measuring these aspects of the body, can be potentially used to identify and prevent symmetry. Without this, symmetry can result in inefficient movement and a disruption of biomechanical balance, leading to increased risk of injury.

After data analysis, a mean inter-limb difference of between 20.98° to 37.28° was found, resulting in $p = 0.5844 - 0.8866$. As this range of values is greater than our threshold value of $p = 0.008833$, null hypothesis 1 fails to be rejected, and no statistically significant difference can be found between left and right ankle dorsiflexion. Shown as a percentage, the mean symmetry percentage calculated is 10.17%. A study targeted towards army rangers revealed those with ankle dorsiflexion differences of $>6.5^\circ$ were 4 times more likely to experience musculoskeletal

injuries, and 5.1 times more likely to experience overuse injury (Almansoof, 2023). While the null hypothesis may have failed to be rejected, it is clear the negative impact limb asymmetry can have on active individuals. With limb asymmetry so frequent across pickleball players, the importance of rehabilitation screening and preventative measures cannot be understated.

6.4 Population Differences

The differences between mean ankle dorsiflexion was also to be compared between differing groups based on sex and age to determine if there was a statistically significant difference that could be found. After data analysis, comparisons were made, and all but one combination (Men 55+ and Women 55+) resulted in a p-value of <0.008833 . This suggests a statistically significant difference between ankle dorsiflexion values across all other age and gender group comparisons, and allows for a rejection of null hypothesis 2. This points to young men and women to have significant differences in ankle flexion, however as time passes the average dorsiflexion steadily decreases and plateaus at a similar point. This conclusion is also supported by literature, citing similar findings of convergence of ankle dorsiflexion over time (Vandervoort, 1992). While it was found that Women had a more significant decrease overall between participants, both populations experienced a roughly 8° decrease between middle to older ages. While dorsiflexion does show trends of decreasing with age, improvements can be made by strengthening weaker dorsiflexion muscles.

Chapter 7: Conclusion

7.1 Broader Impacts Overview

This chapter examines the broader implications of the ankle monitoring device for pickleball players and beyond. Throughout the development of this project, impacts related to engineering ethics, societal benefits, environmental considerations, applicable standards, and economic factors were considered. The following sections reflect the ongoing discussions for the potential effects of this technology as it transitions from prototype to real-world use.

The device meets all design requirements that were set at the beginning of the design process. The device is durable, comfortable, and measures ankle loaded dorsiflexion. It also is a multi-pod setup and includes breathable skin contact straps.

The device can also be used by physical therapists or athletes in other sports. Currently, most physical therapists use goniometers to measure ankle loaded dorsiflexion. A goniometer is a protractor like instrument, which means there can be a lot of user variability. This device can eliminate that variability and give physical therapists a better way to measure ROM. It can also be used by people who play other sports like basketball or gymnastics. Like pickleball, most sports have an emphasis on ankle stability because of lunging, backpedaling or even landing jumps.

7.2 Engineering Ethics

Throughout this project, the team followed the principles outlined in the Mechanical Engineering Code of Ethics, particularly those emphasizing safety, public welfare, and honest reporting of findings.

The team prioritized safety and welfare by designing a non-invasive, wearable device that poses minimal risk to users during physical activity. All materials used, those being PLA and TPU, are compatible with one another and the environment, and safe when coming into contact with skin during use. The design's lightweight design and force-fit mechanism prevent it from becoming a tripping hazard or interfering with natural movement patterns.

Integrity in research and reporting was upheld by conducting in-depth validation studies comparing the IMU-based measurements against established Kinovea video analysis methods. Any limitations of the device were identified and reported, including sample size constraints and the need for wired connections in the prototype.

For ethical considerations, data privacy must be upheld, and informed consent must be given at all times. The user should be notified and aware that the device collects biomechanical data about an individual's movement and potential injury risks.

7.3 Societal and Global Impacts

Our results suggest that the device works and is intended for real-world applications. Physical therapists can use this setup to replace goniometers, as they are prone to user variability.

The setup can allow physical therapists to track the recovery of an ankle injury by analyzing the patient's loaded ankle dorsiflexion over time. Additionally, they can provide injury prevention strategies depending on the recorded angle and assess the patient's recovery time.

In addition to this, consumers may purchase this product for general health purposes. They may use the device personally to assess ankle health and risk of injury during everyday activities, occupation-specific movements, or potential improvements for more active movements. With few alternative methods to easily collect this data without an experienced professional, this device will be extremely useful.

The device, while mostly safe, can pose a slight risk to users, however small. Those with predisposed health conditions may struggle to equip and operate this device, including older populations and those with ongoing & recurring injury to the ankle, knee, or related area. Other groups, such as those with impaired vision, may require extra assistance operating the device and gathering measurements. An additional assistive device has not yet been produced, however is a future consideration for the project.

7.4 Environmental Impact

The current prototype uses PLA (polylactic acid) and TPU (thermoplastic polyurethane) for 3D printing, both of which have environmental implications. PLA is made from renewable resources such as corn starch and sugar cane. It is biodegradable under industrial composting conditions. It has a lower carbon footprint than petroleum-based plastics. The one drawback of the material is that it is not biodegradable in typical landfills or marine environments. TPU is a petroleum-based polymer with a higher environmental impact. It is durable and reusable and can be recycled through specialized programs.

The electronic components within the device, which are the BNO085 IMU and Feather Sense microcontroller, contain rare earth elements and heavy metals that pose environmental risks if not disposed of properly. These include copper, gold, and other metals in the circuit boards.

To promote long-term sustainability, the device can have a return policy for the devices at the end of their life. Its modular design allows for component replacement rather than the device needing to be replaced entirely. Having clear recycling instructions can improve the odds that proper care is taken when discarding the device. Additionally, improving the device for longer use will improve its overall sustainability.

7.5 Codes and Standards

With the intention of the device being tailored to medical usage, both clinical and commercial, certain codes and standards must be followed. Adherence to such regulations ensures efficient and safe production of devices, as well as supporting safe operation in all intended settings.

ISO 13485 Certification is strongly suggested and often required by many global jurisdictions. Targeted at companies engaged in the development, manufacturing, installation,

and servicing of medical devices, it is known for setting quality management standards. Specifically, ISO 13485 looks to ensure companies can design, develop, and deliver medical devices that both meet customer needs and follow all regulatory requirements. Both of these goals need to be met for our design to provide accurate readings consistently, with minimal risk. As such, a strong adherence is required.

ISO 10993 Certification focuses on ensuring strong biocompatibility between manufactured medical devices and consumers. Renditions of this certification have been made recently, with a now required focus on obtaining a strong understanding of potential toxic human interactions before any biological testing. As our devices are constructed of predominantly Thermoplastic Polyurethane and Polylactic Acid, they can be considered safe for usage, as both are generally deemed biocompatible for skin contact, further testing would be required before commercial use regarding the electronics and velcro material.

FDA 21 CFR Part 820, dubbed Quality Management System Regulation, defines good manufacturing practices involving the design, manufacturing, servicing, labeling, storing, packaging and installation of any device intended for human use. Strict adherence must be followed for any Medtech companies within the United States, and as such, our device must fall within the lines listed.

7.6 Economic Factors

Overall, there are minimal economic factors to be considered for the designed device. Cost estimates place the device around \$80 - \$100, taking into account:

- 3-D Printing Filament (TPU/PLA) ~\$20 (bulk)
- Inertial Measurement Unit ~\$20
- Featherboard connection ~\$50
- Velcro straps ~\$9 (bulk)
- Wire connections/Solder/Misc - ~\$5 (bulk)

With the design allowing for material purchases in bulk as well as durability and sustainability, costs can remain low for the manufacturer as well as the consumer. There are currently few devices on the market that serve to accurately measure ankle dorsiflexion, the most common being goniometers, used by Physical Therapists to record flexion on patients for recovery purposes. With higher-quality goniometers costing upwards of \$60, the Ankle Cuff would serve as a digital, slightly more expensive alternative for users who may struggle with manual measurement.

For the intended purposes of prospective medical devices, the broader impacts of the device must also be considered. Without careful reflection, unforeseen issues may arise that could have significant consequences when the device is designed, tested, and used by consumers. Ethically, socially, environmentally, and medically, careful consideration is required to ensure safe and successful implementation.

7.7 Future Work

Due to the nature of the project there are several areas of improvement the team has recognized throughout the process. A more complete and in depth study can and should be considered with these recommendations.

Firstly a common issue experienced during the method of data collection, was the connection between the provided IMU reading ankle loaded dorsiflexion to displaying on a monitor. The original intention utilized a bluetooth connection between the featherboard housed within the heel cuff and a linked monitor, however after researching the required methods to do so, the team quickly realized they didn't have the necessary experience. While possible, it was ruled out to focus efforts in more practical areas, leaving the connection wired between cuff and computer, effectively limiting the freedom of movement of users.

With an increased working time a bluetooth connection could have been further explored. In addition to this, an increase of time would allow for a much larger sample size. At project end, a total of 60 participants have been recorded, however ideally the number would have been much higher for a more representative view of the population.

Finally, project funding served to restrict the quality of sensors purchased and methods of measurement. With an increase in allocated funds, more accurate sensors would have been considered, as well as the potential for acquisition of an easier method to transfer data from featherboard to laptop, omitting a wired connection.

Overall the project has served to be successful, however there is always room for improvement. It is recommended to any that seek to replicate or attempt the procedure or any similar from this study to consider these limitations, either mitigating their presence or planning around them.

Moving forward from the project, several steps can and should be taken. Additional participants should be tested to better represent the overall population of pickleball players, including targeting different skill levels, injury status / history, location, and age. Movements more reflective of those done during play should also be replicated and recorded to gain a more relevant flexion. Additional sensors can be added or exchanged for more accurate readings of flexion, or readings of both ankles simultaneously, as well as recorded for a longer span of time to record more long-term flexion effects / the development thereof. With this information, injury risk can be predicted according to the factors recorded, with recommendations for injury prevention to be developed.

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Appendices

Appendix A: Participant CV & Symmetry Metrics

Below is the coefficient of variation of ankle dorsiflexion ROM across five trails for each participant for a given sample group. There is also inter-limb symmetry for each participant within a sample group. Both these data sets were used to determine average CV and inter-limb symmetry for a given group.

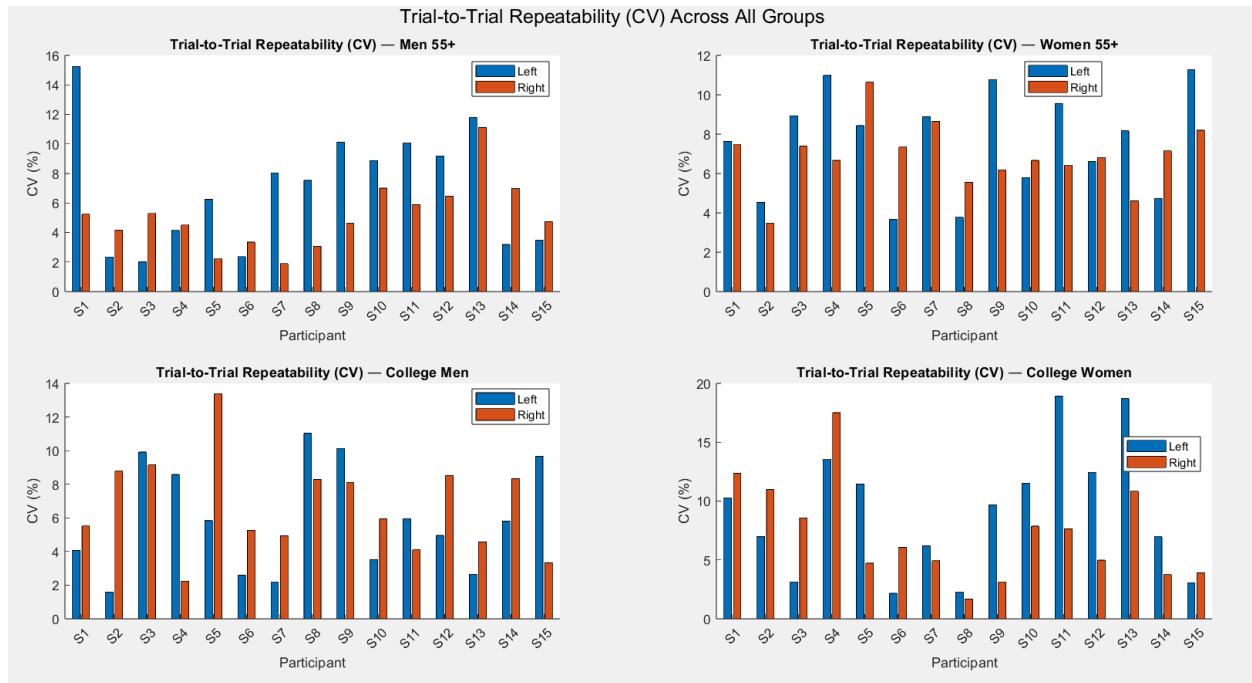


Figure A.1: Coefficient of variation (CV) of ankle dorsiflexion ROM across five trials per ankle for each participant. Lower CV values indicate greater repeatability.

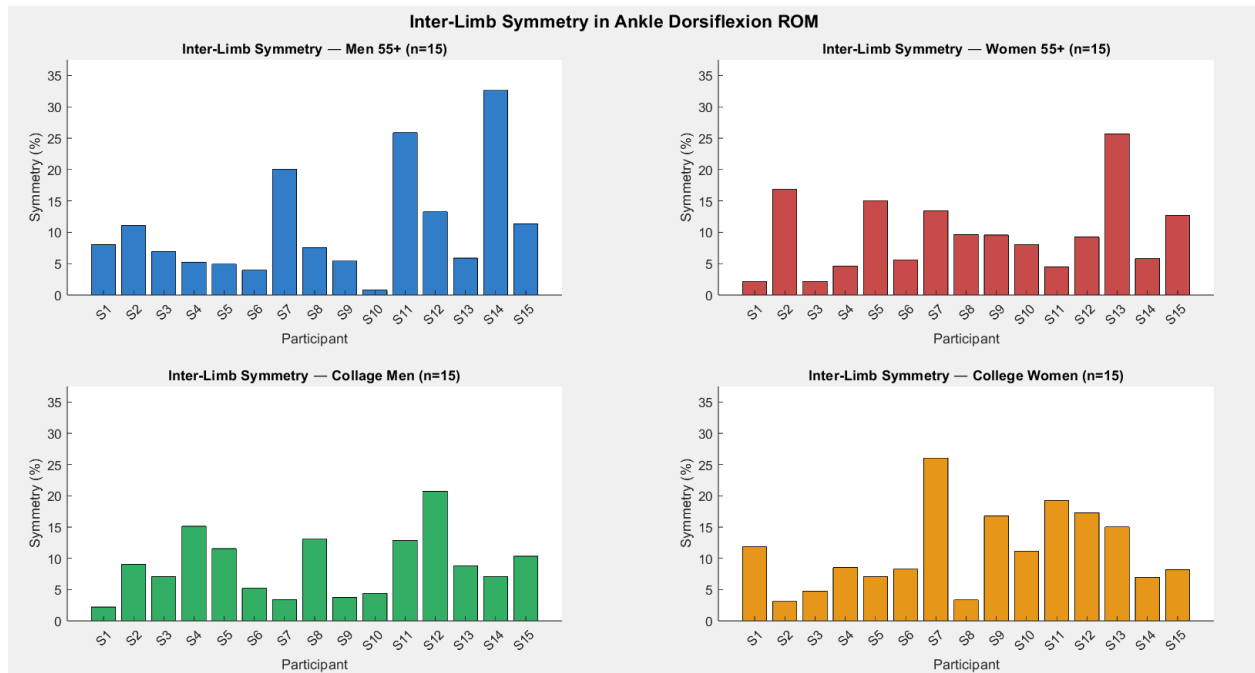


Figure A.2: Inter-limb symmetry in ankle dorsiflexion ROM expressed as percentage difference between the two ankles for all four groups